eLife

Reviewed Preprint
v2 • November 7, 2025
Revised by authors

Reviewed Preprint
v1 • July 23, 2025

Neuroscience

Formation of Task Representations and Replay in Mouse Medial Prefrontal Cortex

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eLife Assessment

This **useful** study characterizes the evolution of medial prefrontal cortex activity during the learning of an odor-based choice task. The evidence provided is **solid**, providing quantification of functional classes of cells over the course of learning using the longitudinal calcium recordings in prefrontal cortex, and quantification of prefrontal sequences. However, the experimental design appears to provide limited evidence to support strong conclusions regarding pre-existing representations or the functional relevance of neural sequences. The study will be of interest to neuroscientists investigating learning and decision-making processes.

https://doi.org/10.7554/eLife.106981.2.sa4

Abstract

The medial prefrontal cortex (mPFC) is thought to support cognitive flexibility by forming and maintaining generalized representations of abstract tasks. The formation of these representations as well as their relation to preexisting representations of contextual or spatial information is incompletely understood. In this study, we analyzed longitudinal 1-photon calcium recordings in mice performing an olfaction-guided spatial memory task over an eight-week period that included habituation, training, and sleep epochs. Our results reveal that, while a minority of neurons initially conveyed significant information about the behavior of the animal, the bulk of task-related activity only emerged after the animals reached proficient performance. Although goal arm information is robustly represented at both the single-cell and network levels both during learning and in task proficient mice, it undergoes significant remapping throughout the learning process. Additionally, we identified the establishment of recurring sequences during learning and their replay at reward locations, with no evidence of them existing during odor sampling phase, during sleep or before training. Conversely, during odor sampling, information about odor identity is robustly available in the rate coactivation patterns, even before animals reached task proficiency. These findings suggest that the mPFC predominantly establishes generalized task

representations de novo during learning, relying only minimally on preexisting spatial representations and that sub-second neural sequences in the mPFC are more likely involved in evaluating behavioral outcomes rather than planning future actions.

Introduction

The medial prefrontal cortex (mPFC) is generally implicated in the representation of abstract task variables [Miller & Cohen, 2001; Brincat et al., 2018, Kaefer et al., 2020; Tang et al., 2023; El-Gaby et al., 2024] as required e.g., in rule learning [Wallis et al., 2001; Goodwin et al., 2012; Reinert et al., 2020]. In depth understanding of neuronal activity patterns associated with abstract tasks and their establishment during learning strongly profits from experimental approaches available in the rodent system. However, in addition to anatomical differences between primates and rodents [Laubach et al., 2018; Preuss & Wise, 2022; Hanganu-Opatz et al., 2023], behavioral task designs for rodents are also often affected by spatial components, which introduce correlations between task phases and spatial locations [Sauer et al., 2022; Muysers et al., 2024; Muysers et al., 2025]. On the one hand, this interrelation between space and task complicates interpretation of the results, on the other hand, it, however, also provides an opportunity to study the effects of spatial schemas to task learning, as hypothesized from human fMRI data [Zheng et al., 2021]. Unlike in the hippocampus, where CA1 place cells undergo global remapping in different contexts, PFC representations have been found to remain stable across environments, allowing for task generalization [Tang et al., 2023, Muysers et al., 2024]. This stability suggests that new experiences do not completely overwrite PFC representations but instead integrates new information into existing structures, making the PFC a hub for cognitive flexibility. However, the precise nature of this reorganization remains debated. One perspective suggests that task representations in the PFC emerge de novo during learning and are distinct from preexisting spatial representations [Symanski et al., 2022]. This view is supported by findings that task-selective sequences arise only after learning, indicating a learning-driven transformation of PFC activity. An alternative perspective, however, argues that PFC task representations build upon preexisting spatial schemas rather than forming them entirely anew [Zheng et al., 2021]. Accordingly, the fact that PFC representations remain stable across environments may suggest that learning-related changes reflect adaptations of existing networks rather than the formation of entirely new assemblies within a circuit [Tang et al., 2021; Nardin et al., 2021]. Whether PFC task representations are reorganized from prior spatial schemas or develop independently remains unresolved, highlighting a fundamental gap in our understanding of cortical learning mechanisms.

In this study, we set out to study the development of task representations on longitudinal 1-photon recordings and to test whether they are based on preexisting spatial scaffolds. We reasoned that, in naive adult animals, task representations should only develop over learning, whereas the concept of space should be readily available during or even before training. We further addressed the question, whether task representations emerge as sequential assemblies in the mPFC network, or whether they are reflected as rate coactivation patterns. The former hypothesis would imply that pair correlations or activity sequences in neural populations could be interpreted as replay/preplay , whereas the latter would imply that aggregated task information would be available in population patterns and potentially provide a substrate for working memory.

Making use of a partly previously published dataset containing longitudinal mPFC recordings [Muysers et al., 2024, Muysers et al., 2025] over up to 8 weeks, encompassing habituation, training and sleep periods, we were able to explore the development of the task code during learning. We found a small subset of goal arm selective cells that develop general task selectivity and maintain the location of the place field. However, most of the task-related activity was observed only when the animals were proficient in task performance. Moreover, we found goal arm-selective information to be more strongly represented than general task phase information, both on the single cell and the population level. A small fraction of goal arm-selective cells showed stable firing fields throughout the recording period even before learning has started. Establishing sequence analysis on calcium transients obtained from 1-p imaging, we identified the emergence of generalized task-selective sequences after learning, whereas we could find no evidence for task-related preplay during sleep or habituation to the arena. Both findings suggest that the mPFC task code is established during learning. Thus, mPFC task representation seems to be established during learning with only a small amount of stable spatial tuning that is unaffected by learning.

Results

Mice learned to perform an odor-guided navigation task (**Fig. 1A**), while calcium activity was imaged in the mPFC using a miniscope [Muysers et al., 2024]. Depending on odor identity, mice had to learn to choose either the left or the right arm and were rewarded upon the correct choice. From previous analyses [Muysers et al., 2024, Muysers et al., 2025], we expected the recorded population to display place-specific activity and asked how many of the cells were significantly spatially informative after animals became proficient in the task. We employed a complementary approach to [Muysers et al., 2025] by testing for significance of spatial information content [Skaggs et al., 1993] (rate maps computed separately for left and right arm) with shuffle controls (see Methods), and only included data from outward runs. We found that about half of the cells were significantly informative (SI) about the goal-arm-specific location on the track. The place maps of SI cells, separately extracted from odd and even trials in the maze confirmed the statistical approach (Example animal in **Fig. 1B**). Since in this task, space is highly correlated with the task phase, we set out to distinguish place coding from task coding first, by training a Bayesian decoder on the whole population of recorded cells to predict goal-arm-specific spatial position based on the recorded activity (**Fig. 1C**). We found that the decoding error obtained by cross decoding (decoding left goal trajectories from the decoder trained on right goal trajectories and vice versa) had a similar magnitude as the error obtained from training and decoding on subsets of trials with the same goal arm (**Fig. 1D**). Both errors were considerably and significantly lower than those obtained from shuffle controls ($p<0.02$ for both errors; permutation test, 50 shuffles). To further corroborate the evidence that mPFC place coding has a high contribution of task-selective activity, we differentiated between goal-arm-selective and generalized task selective cells, computed rate maps separately for left- and rightward trials of SI cells (**Fig. 1E**) and computed correlation coefficients. Those SI cells for which the correlation exceeded the 95%-tile of the shuffle distribution (Methods) were called generalized task phase selective cells (TCs), whereas SI cells for which the two maps were not significantly similar were called goal-arm selective cells (GCs). We found a ratio of about 48% of TCs among SI cells (**Fig. 1F**) over all animals, considering only the “learned” sessions in which the animals have reached a performance criterion described previously (Methods; Muysers et al. (2024)).

In previous analyses generalized task-selectivity on the population level was shown to increase over learning (Muysers et al., 2025), we therefore examined how goal arm- and task phase-selectivity of SI cells develops over learning (**Fig. 2**). We therefore subdivided sessions into early “learning” sessions and late “learned” sessions based on a behavioral performance criterion of 70 % correct trials (Methods; Supplement to **Figure 1A-C**). During learning sessions animals had a strong bias towards left turns (**Fig. 2A**; Supplement 1 to **Fig. 2A**), which was also reflected in the over proportionate number of SI cells with fields in leftward trials (**Fig. 2B**; Supplement 1 **Fig 2B,C**). After learning, both the behavioral and the neural biases vanished (**Fig. 2A,B**). New SI cells were predominantly added for rightward trials (**Fig. 2B**), whereas about a third of the SI cells for leftward trials did not remain SI in the learned condition (**Fig. 2B**). We then distinguished learning-induced changes between goal arm and task phase encoding cells and found that in learning trials there was a strong bias towards goal arm encoding (**Fig. 2C**), whereas in the learned condition a substantial fraction of TCs (about 40%) was consistently found in all animals (**Fig 2C**). The increase in generalized task phase encoding on the level of SI cells

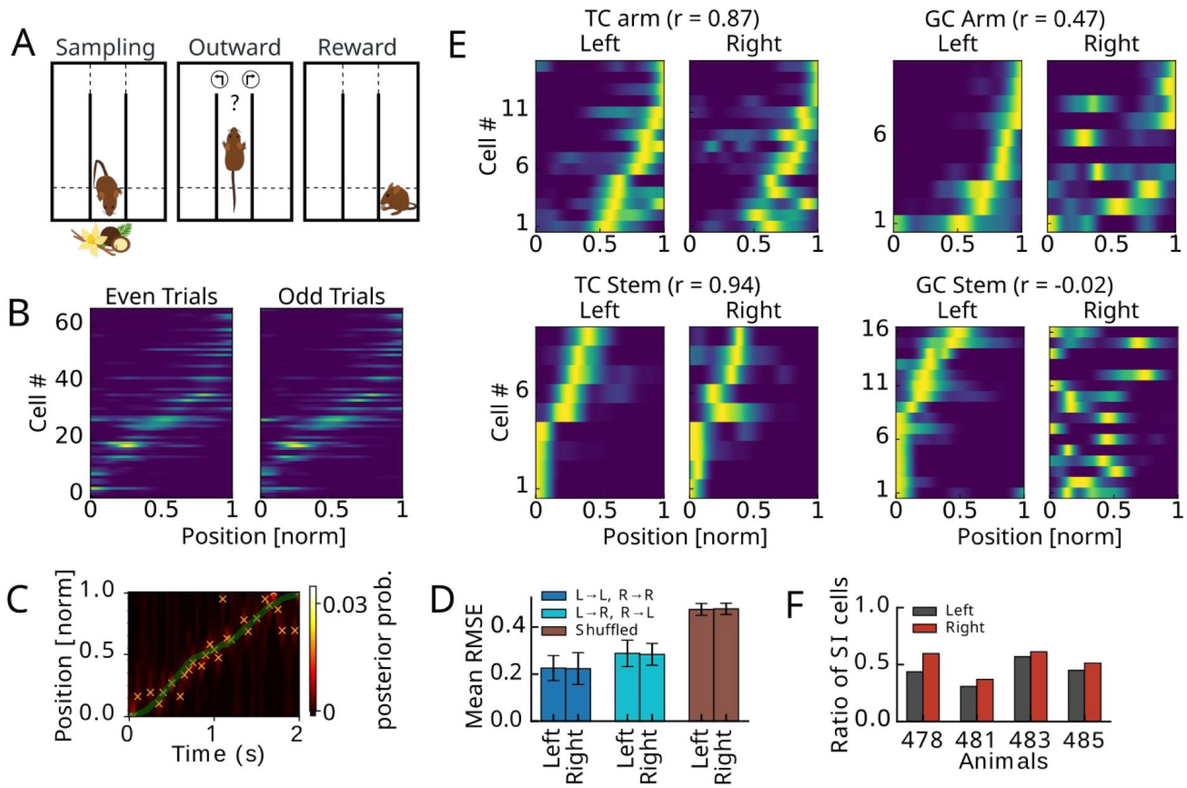


Figure 1.

Spatially informative mPFC principal cells from proficient mice performing an odor guided navigation task.

(A) Mice were performing an odor guided navigation task. We separated analysis into three behavioral phases. During Sampling animals were exposed to an odor. During Outward the animals were moving on the stem and one of the arms (above dashed line). During Reward the animals were in the reward area (below dashed line). (B) Activity maps (computed separately for left and right arm) obtained from 1-photon calcium imaging were tested on significant spatial information (SI) content. Maps of cells with significant SI (example session) are matching between even and odd trials. (C) Example trial in which maximum posterior decoding (crosses) is aligned with true position (green). (D) Root mean square decoding errors (RMSE) for all animals and all session separated for left and right arm trajectories. Cross decoding RMSE (light blue) is still significantly lower than RMSE from shuffling ($p<0.02$; none in 50 cell index shuffles), indicating a large contribution of left-right invariant, i.e. task phase selective activity. (E) Recordings from an example animal 485 separated into generalized task phase selective cells (left) and goal arm selective cells (right) with left and right activity maps significantly correlated or not. (F) About half of the cells are significantly spatially informative in all animals.

matches the increase of task phase encoding previously observed on the population level in an extended population of animals (path-equivalent cells in [Muysers et al., 2025]). Despite the growth of the TC population, GCs remained to be the majority of cells, particularly those GCs with a field in the stem (**Fig. 2C**). GCs on the stem have been previously referred to as splitter cells in the hippocampus [Wood et al., 2000] and suggested to encode episodic information. This is in line with our interpretation of GCs as goal encoding cells. Also only about 40% of initial TCs remained TCs after learning (**Fig. 2C,D**). The relative reduction in GCs, the increase in TCs and the instability of a fraction of TCs over learning indicates that emerging TCs in the learned condition may have been generated from previous GCs. Indeed ~40% of the TCs in the learned condition arose from GCs, however, also a similar fraction were not SIs during the learning condition (**Fig. 2C,E**). For GCs that became TCs in the learned condition, the majority kept their place field (in an interval +/-0.1 relative to the original field center; see example place cell population in **Fig. 2F** and summary statistics in **Fig. 2G**). Thus, task coding develops over learning by addition of existing SI cells with already formed place fields as well as the acquisition of TCs from previously non-spatially tuned cells. However, although most of the selectivity arises over learning, a small but significant fraction of cells show stable rate maps not only in comparison between learning and learned conditions (**Fig. 2G**) but also taking into account the habituation phase, where the animals were exploring the behavioral arena without a task (Supplement 2 to **Figure 2**). A significant fraction of stable rate maps (habituation vs. learned) could particularly be observed in TCs but not in GCs (Supplement 2 to **Figure 2F**). This finding is consistent with the hypothesis of a limited preexisting scaffold.

We next reasoned that the establishment of a task code on the single cell level during learning will likely be associated with changes in covariance structure on the network level. Since a subpopulation of TCs inherited their firing fields from previous place fields, we reasoned that these initial firing properties could reflect the activity of previously existing cell ensembles. In a first set of analyses we were testing whether the rate covariance structure during the outward phase is “reactivated” in the sampling and reward phases using the reactivation measure proposed in [Peyrache et al., 2009; Peyrache et al., 2010] (**Fig. 3**). Both during learning and learned phase reactivation during sampling was significantly larger than during the reward phase (Wilcoxon test, Learning: p=0.016, rank=1, n=8 sessions, Learned: p=1.2e-10, rank=0, n=34 sessions), arguing that already during sampling cell assemblies active during outward running start to co-fire. Moreover, comparing reactivation of left or rightward patterns during sampling and reward phases in correct trials between a true goal (patterns were taken from the trials with fitting goal arm) and a swap goal (patterns were taken from the trials with opposite goal arm), reveals significantly stronger reactivation in the sampling period for the “true goal”, both for learning and learned condition (Wilcoxon test; Learning: p=0.040, rank=3, n=8 sessions; Learned: p=2.4e-7, rank=30, n=34 sessions). These findings provide strong evidence that the cell assemblies during sampling are goal arm specific, i.e., could underlie a sensory-driven recall initiated by odor sampling. For the reward phase, reactivation reached significant goal arm specific only in the learned condition (Wilcoxon test; Learning: p=0.11, rank=6, n=8 sessions; Learned: p=7.2e-6, rank=56, n=34 sessions). Reactivation was significantly larger during sampling vs. reward even in the swapped condition (Wilcoxon test, Learning: p=0.032, rank=3, n=8 sessions, Learned: p=1.2e-5, rank=42, n=34 sessions), and reactivation strengths were not significantly different between learning and learned condition (Ranksum test; sampling: p=0.53, z=-0.64, n1=8, n2=34 sessions, reward: p=0.64, z=0.48, n1=8, n2=34 sessions; not shown in Figure), suggesting that the co-firing structure reflects olfactory cue rather than task information [Taxidis et al., 2020; Sun et al., 2024].

In addition to rate covariance (co-firing) based assemblies, mnemonic processes have also been associated with temporal activity pattern, i.e., activity sequences, particularly in the hippocampus.

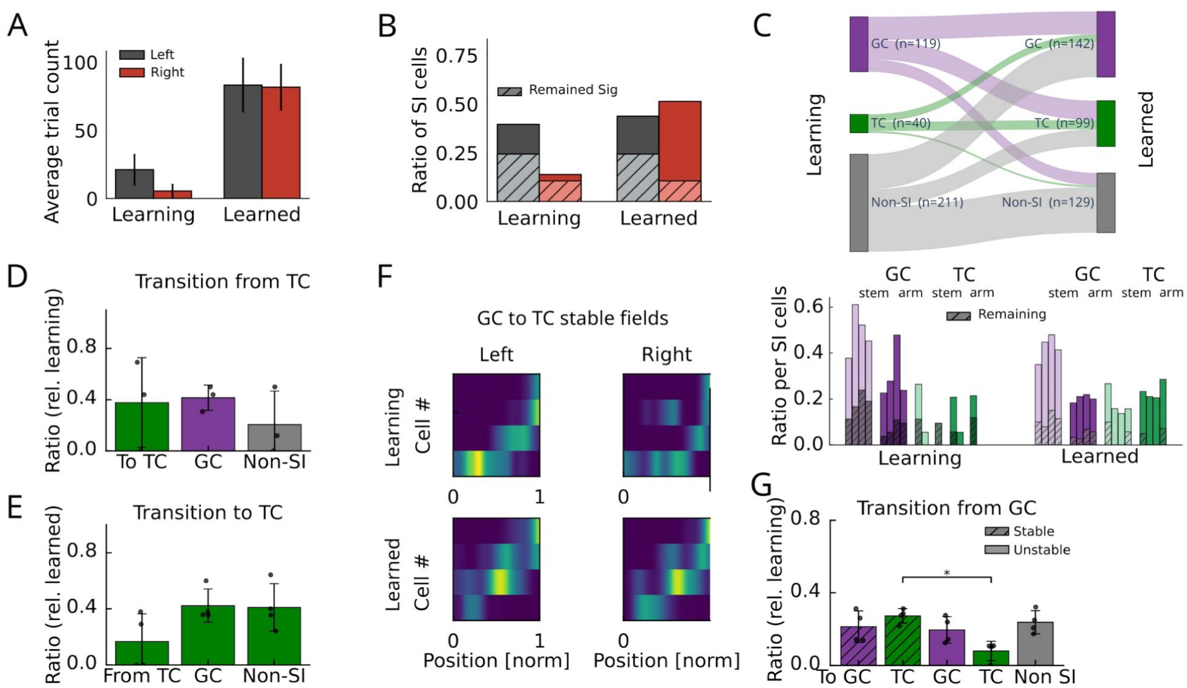


Figure 2.

Development of task tuning during learning.

(A) Trial counts (mean and standard deviation) over all animals in the learning and learned condition. Recordings in both conditions were limited to 15 minutes, explaining the lower number of trials in the learning group, where animals are still familiarized with the task structure. (B) Fraction of SI cell averaged across animals in the learning and learned condition. Hatched areas indicate proportions significant in both conditions (overlapping cells). (C) Top: Transition between different cell types between learning and learned sessions. Bottom: Amount of GCs and TCs normalized by the number of SI cells for all 4 animals (individual bars) during learning and learned condition. Hatched areas indicate proportions of cells that remain in the same category after learning. (D) Fractions of task cells (during learning) that transitioned to TC, GC and non-SI cells after learning. (E) Fraction of task cells (for the learned condition) that arose from TC, GC and non-SI cells during learning. (F) Example of GCs during learning (in one animal) that became TCs after learning. (G) Fractions of GCs (during learning) that transitioned to TC, GC and non-SI cells with stable (hatched) and non-stable (plain) place field location. A Wilcoxon signed rank test showed significantly larger ($p = 0.028$, $U=0$, $N=4$ animals) stable TCs than non-stable TCs.

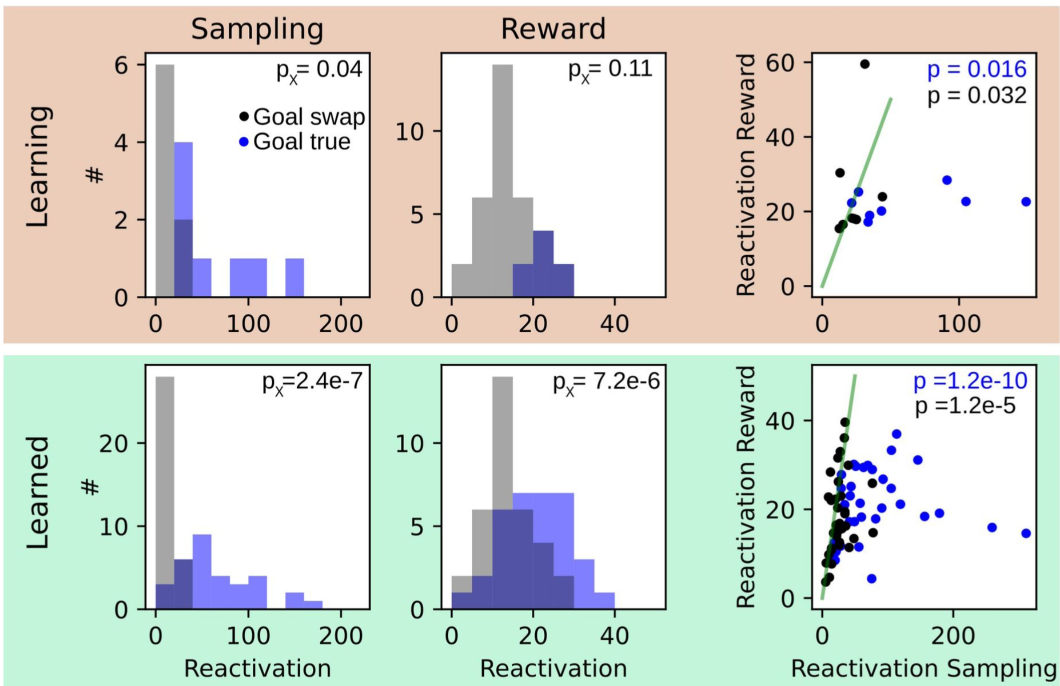


Figure 3.

Reactivation of cofiring patters is goal arm specific.

Both, during the learning (top, orange) and the learned (bottom, green) condition, we computed the reactivation strength [Peyrache et al., 2010] of patterns observed during outward running for the sampling and the reward phase. For each session, reactivation was computed with respect to the true trial label (blue) and a swapped trial label (black). Left: For the true label reactivations were significantly (p_X) larger than for swapped labels expect for reward phase during learning. Right: Reactivation during sampling was significantly (p) larger than during reward for true trial labels (identity is illustrated by green solid line). P values were derived from Wilcoxon signed rank tests. Statistics are given in the main text.

We therefore next applied an unbiased sequence detection algorithm (Chenani et al., 2019) based on z-scored rank order correlations. In order to select only time points with many concurrently active cells, we set a threshold to the population calcium traces (Methods) and extracted time intervals of 500 ms centered around the peaks of the population trace (**Fig. 4A**). Burst rates were significantly larger during behavioral epochs (covering about 20 to 40 % of the data; Supplement to **Figure 1** D, E) than during sleep and during periods of habituation to the arena (Task vs. sleep: $p = 3.1 \times 10^{-130}$, U = 4,974,538, n=7390 ; task vs. arena: $p = 2.6 \times 10^{-5}$, U = 1,546,011, n=6732 ; Mann-Whitney U rank tests; **Fig. 4B**), already indicating limited opportunities for preplay (i.e. behaviorally correlated sequences occurring before the start of the outward phase). Moreover, burst rates during correct trials were significantly lower than during error trials (p = 0.0014, U = 3461962, n=6292 ; Mann-Whitney U rank test; **Fig. 4B**). Ensembles were then identified according to the order of their center of mass calcium activity during population bursts (**Fig 4C** and Supplement 1 to **Fig. 4**). Hierarchical clustering of the z-scored correlations was done longitudinally over all behavioral sessions during the learning and learned conditions based on similarity scores computed for all pairs of bursts in these times frames (Methods). With this approach we were able to identify repetitive activity sequences over the whole course of the experiment (see Rastermap plots [Stringer et al., 2019] **Fig 4D**; Supplement 2 to **Fig. 4**). Visual inspection of the Rastermap plots and correlating time points of sequence activations with behavioral trajectories exhibited large differences of sequence activation between learning and learned condition, both in the spatial patterns (example animal in **Fig. 4D**) and the distribution of the sequences (**Fig. 4D,E**). Rastermap plots (**Fig. 4D**) also reveal little similarity of sequence expression between task and habituation or sleep condition. Sequence expression during the task was more variable between learning and learned condition than between correct and error trials (**Fig. 4E,F**) as revealed from shuffled cluster identities (P-values were obtained using a permutation test with 200 shuffles: learning vs. learned, correct: $p < 0.005$; failed: $p < 0.005$; correct vs. failed, learning: p = 0.8350; learned: p = 0.415). Thus, our data suggest that task learning is associated with the establishment of a new robust population representation of current behavioral states.

In order to find functional correlates of the recurring sequence activity, we again computed SI values of sequence clusters and found a significant increase in SI clusters over learning (Wilcoxon signed rank test, p = 0.02, W=1, n = 8; **Fig. 5A**; Supplement 1 to **Fig. 5A,B**), similar to the increase in SI cells reported in **Fig. 2B**. Separating goal arm selective sequences (GS) from generalized task phase selective sequences (TS) by applying the same approach as for GCs and TCs (see Methods), we found an increase of both GSs and TSs over learning with a particular bias to goal specificity in the arm (**Fig. 5B**), different from the stem bias of GCs reported in **Fig. 2C**. The majority of sequence activity, however, was neither related to TSs nor to GSs (**Fig. 5C**). The only exception was during the outward runs, where GSs and TSs together contributed to sequence activity by about as much as non-SI sequences. To understand the origin of goal and generalized task phase tuning of sequence clusters we then examined the place fields of the contributing cells and found that particularly place fields in TCs had a stronger contribution to sequence activity in outward runs (**Fig. 5D**). Moreover, to link place fields of SI cells to fields of SI sequence clusters, we computed distributions of distances between the peak of the cluster field and place fields of the contributing cells (pooled histogram shown in **Fig 5E**; Histograms for individual animals in Supplement 1 to **Fig 5C**) and tested for significance, by comparing the distribution of peak distances between all cluster and cell fields. For all 7 detected TSs the field centers of the TCs significantly determined task tuning of the sequence clusters, whereas for the 13 GSs only less than the half were determined by the place field centers of GCs (**Fig. 5F**). Thus, there is a stronger link between TCs and TSs as compared to the link between GCs and GSs (p = 0.0047; Fisher's exact test).

Taken together, our findings suggest that recurring sequences during outward runs are largely supported by TCs, however, the overall frequency of their occurrence, and potentially their functional relevance for task representation is rather limited. We therefore were wondering,

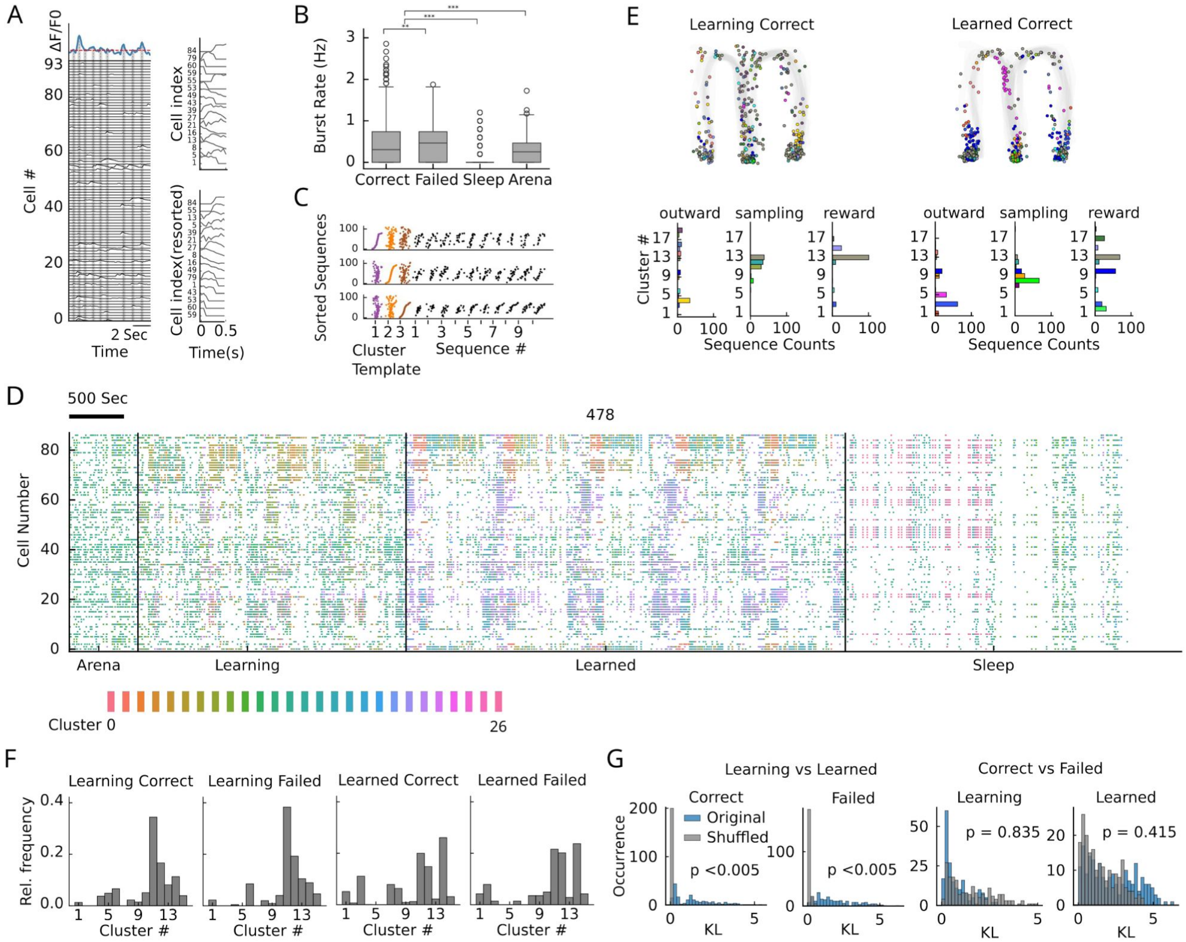


Figure 4.

Recurring sequence motifs in 1-photon recordings change with learning.

(A) Example population recording of 93 cells. Blue trace indicates population (sum) activity. Significant peaks (above red line) of population activity are identified as bursts. Within a 500 ms around burst peaks, cells are sorted according to the center of mass of the calcium trace. The index sorting is called a sequence. (B) Burst rates combined over all animals for different behavioral states. Correct and Failed correspond to Sampling, Outward and Reward from **Fig. 1A**, Sleep is derived from intermittent sleep epochs (see Methods), Arena denotes the habituation period where the animal was left in the arena without task (see Methods). Significance is derived from Mann-Whitney U rank tests (see main text). (C) Three example sequence cluster templates (colored) obtained via hierarchical clustering (see Methods) sorted according to first, second, and third cluster. Black sequences are representative cluster members of the cluster for which the cell index sorting was made in the panel. (D) Rastermap visualization [Supp. Figure 3 A; and Stringer et al., 2019] of sequences clusters (colors) for an example dataset from a single animal (478) for all sessions concatenated and sorted according to behavioral condition (Arena habituation, in task during Learning condition, in task during Learned condition, Sleep). Each point represents the time a cell had a calcium transient, with colors indicating the active sequence cluster. Sleep sessions were dispersed throughout but shown here as a concatenated block for visualization purposes. (E, Top) Cluster identity (color) of a populations burst as a function of position on the maze in the learning and learned condition. (E, Bottom) Distributions of cluster identities for all burst in the behavioral phases. (F) Relative frequencies of cluster identities of bursts in one example animal during learning and in the learned condition subdivided for correct and failed trials. (G) Kullback-Leibler (KL) divergence of cluster distributions (as in panel F) for all animals (blue) were obtained by subsampling the number of bursts in the learned conditions to match the number of bursts in the learning conditions (see Methods). Animal-wise shuffles of learning and learned labels in the subsampled histograms are plotted in gray.

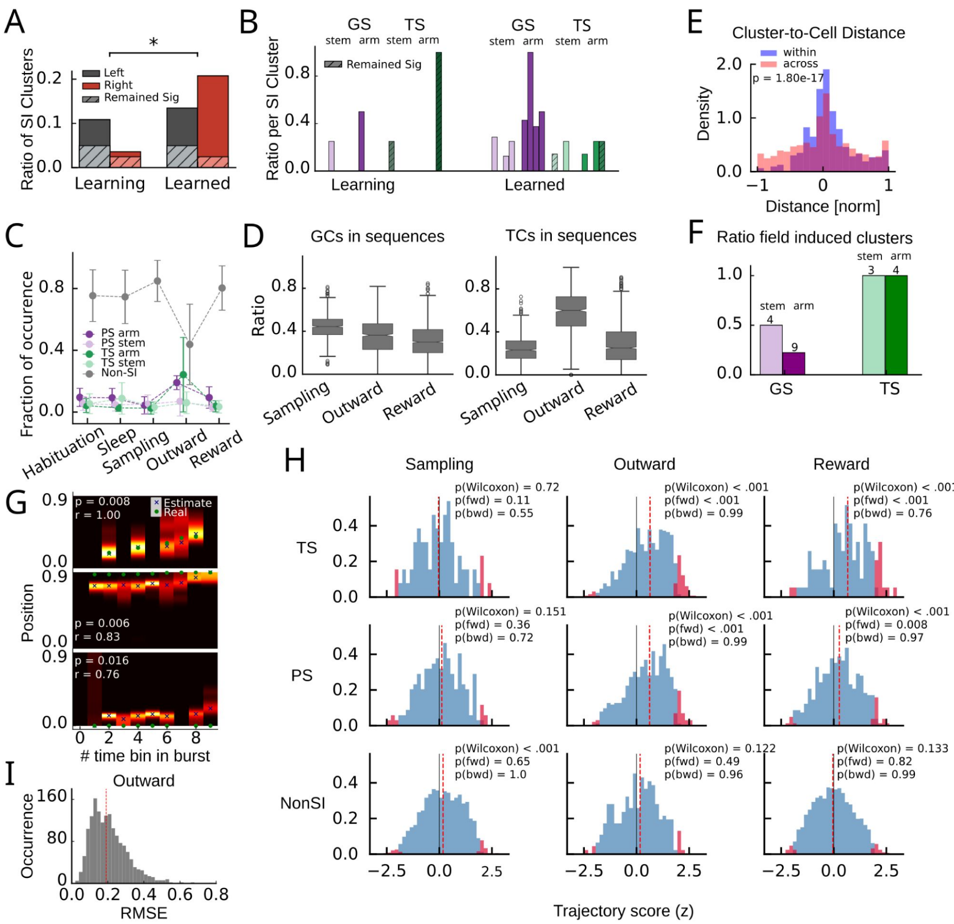


Figure 5.

Generalized task phase- and goal arm selective sequences are used for replay but not planning.

(A) Fraction of SI sequence clusters (relative to all clusters) combined across animals in the learning and learned condition. Hatched areas indicate proportions significant in both conditions. Significance is tested according to a Wilcoxon signed rank test (p=0.02). (B) Amount of GS and TS clusters normalized by the number of SI clusters for all 4 animals (individual bars) during learning and learned condition. Hatched areas indicate proportions of clusters remaining in the same category after learning. (C) Fraction of bursts attributed to GS, TS and non-SI sequence clusters during different behavioral phases. The majority of bursts is not selective to behavioral parameters except in the outward phase, where about half of the bursts are associated with behavior. (D) Fractions of GCs and TCs contributing to bursts of the different cluster types and different behavioral phases. (E) Histogram of distances between field peaks of the cluster and the field peaks of the contributing cells for all SI clusters (blue). Distances between field peaks of all SI cells and all sequence clusters (across cluster) are shown in red. Clusters are called significantly cell induced if their absolute across cluster distances are significantly larger than the within cluster differences according to a one-sided Mann-Whitney U rank test. (F) Fraction of cell induced clusters (see panel E) during the learned condition. Number on top of bars indicate total numbers of clusters of that type over all animals. (G) Examples of decoding during bursts. Posterior is color coded (max normalized). Crosses indicate maximum posterior, green dots the true position of the animal in the respective time bin (50 ms). Top: Example during outward phase; Middle: example during reward phase; Bottom: example from odor sampling phase. (H) Distribution of z-scored rank order correlation coefficients between time bin and decoded position (trajectory scores) in each burst during sampling (left column), outward (middle column) and reward (right column) phase for generalized task phase-selective (top row), goal arm-selective (middle row) and non-SI (bottom row) clusters. Z-Scoring was performed using mean and standard deviation of a distribution obtained from randomly shuffling cell indices. Red bars indicate significant forward and backward replay for which trajectory score exceeded the upper 97.5 or the lower 2.5 percentile of the distribution generated by cell id randomization in the respective session. The fractions of forward and backward replay were then tested for significance above chance (0.025) according to a binomial test; p(fwd) & p(bwd). Red dashed lines indicate the medians, p(Wilcoxon) values are derived from a Wilcoxon signed rank test for zero median. (I) Distribution of RMSE from the outward phase over all animals (Red line indicates median).

whether sequence activity correlates with trajectory replay as demonstrated for the hippocampus [Lee & Wilson, 2002]. By applying the Bayesian decoder generated for trajectory decoding from single cells (**Fig 1C**) trained on activity during population bursts (**Fig 5G** and Supplement 2 to **Fig 5** for examples), we found correlations of decoded position with time bin (within a 500 ms burst) strongly exceeding chance level only during outward and reward phase, for both GSs and TSs (**Fig 5H**), with only fractions of significant forward replays exceeding chance levels. We reckoned that strong correlations between decoded position and time during outward phases may simply reflect the behavioral sequence (running trajectory, as in the first example shown in **Fig 5G**). We therefore computed decoding errors during bursts in outward runs (**Fig. 5I**) as root mean square error (RMSE) between true and decoded trajectories. The median RMSE of about 0.2 matches the RMSE from trajectory decoding in **Fig.1D**, supporting the interpretation that significant correlations between decoded position and time during SI sequences in outward runs in **Fig. 5H** reflect the real time task representation. Significant correlations of GSs and TSs in the reward phase, however, cannot relate to the current position of the animal and therefore have to be considered as replay. During the odor sampling phase, no such significant replay was observed for GSs and TSs suggesting these bursts are not involved in odor sampling or even planning, but rather in the execution and evaluation of behavior. Non-SI sequence clusters showed small but significant bias to preplay in the sampling phase, indicative of Non-SI sequence clusters potentially containing a small contribution of spatial subsequences. However, in contrast to GSs and TSs that show significant proportions of forward replay (and not backward replay) in both the outward and reward phases, Non-SI sequences during sampling did not reveal significant forward preplays exceeding chance expectations. Replay was not significantly observable during the reward phase in the learning condition (Supplement 1 to **Fig. 5D**), which suggests that replay requires a stable task representation, or at least extensive exposure to the task. During the outward phase, however, correlations between decoded position and time were significant for all clusters (Supplement 1 to **Fig. 5D**), even for the non-SI clusters, supporting the interpretation of a real time behavioral representation even in the learning condition. Together with our finding of strong changes in sequence expression after learning (**Fig 4E**) these findings suggest that a representation of task develops during learning, however, it is not reflecting sequence structure during learning and habituation.

To conclude, generalized task phase representation by TCs developed in parallel to a goal arm representation by GCs, where the latter is supported by an even larger number of SI neurons (predominantly in the stem of the M-maze, i.e. splitter cells [Wood et al., 2000]) and an even larger amount of GSs (predominantly in the arm of the M-maze) than TSs. Thus, goal and task phase representation seem to coexist in rodent mPFC.

Discussion

Longitudinal measurements of 1-p calcium activity in mice learning and performing an odor-guided navigation task allowed us to analyze the development of goal arm and generalized task phase selectivity on the single cell as well as on the population level. Recordings from animals that have not yet reached good task performance exhibited a preference of goal arm selective neurons as compared to generalized task phase selective neurons. However, also in task proficient animals, with many generalizing TCs, a sizable amount of cells showed goal arm selectivity. GCs, however, largely remapped during learning, whereas TCs developed at least partly from initial SI cells with overlapping place fields, which indicates that task topology could be inherited at least to a small extent from existing spatial schemas. This hypothesis is further corroborated by a small but significant proportion of cells that exhibit a firing field at the same location in the arm even during habituation phase, without performing a task. Extending our analysis to the coordinated activity of neural sequences, we found that learning leads to the emergence of new structured activity patterns, indicative of refined network organization. A large number of recurring activity motifs were present, but TSs almost exclusively arose after learning. These changes were not significantly different between correct and failed trials and suggest a broad reconfiguration of sequence dynamics in the mPFC. These findings support the idea that task representations are generated de novo in the mPFC during learning.

While memory related fast (sub-second) sequence activity/temporally correlated firing has been previously reported in the PFC using electrophysiological methods [Euston et al., 2007; Fujisawa et al., 2008; Peyrache et al., 2009; Kaefer et al., 2020], the present study, to our knowledge, identified for the first time fast recurring neural sequence activity from 1-p calcium data, based on correlation analysis, thereby allowing to study the formation of fast sequences over a period of weeks. Using the center of mass approach for the calcium traces allows to interpolate between time samples and thereby limits the temporal resolution to the inverse of the sampling frequency rather than being constrained by the kinetics of the calcium indicator, at least for moderate sampling rates below about 50 Hertz.

So far, sequences on fast time scales have mostly been reported for the hippocampus during theta oscillations [Foster & Wilson, 2007] as well as during quiet wakefulness [Diba & Buzsaki, 2007; Foster & Wilson, 2006] and sleep [Lee & Wilson, 2002]; but see [Euston et al., 2007; Kaefer et al., 2020]. The functional role of hippocampal sequences is still debated and mostly assumed to relate to memory consolidation [Girardeau et al., 2009; Fernandez-Ruiz et al., 2019] and planning [Kay et al. 2020; Jadhav et al., 2012]. In contrast to hippocampal sequences, we find a very much reduced incidence rate outside the behavioral task, particularly during odor sampling. Odor sampling is a state of active sensing [Martin et al., 2007; Kay, 2014], and olfactory processing in PFC is not only reflected by co-firing patterns (**Fig. 3** and [Taxidis et al., 2020; Sun et al., 2024]), but has also been reported to affect oscillatory activity [Martin and Ravel, 2014], particularly phase amplitude coupling of beta and gamma oscillations to the theta rhythm [Ramirez-Gordillo et al., 2022]. The absence of recurring sequences during the sampling phase are, thus, likely reflecting that active sensing and planning are distinct behavioral states. The absence of replay during sleep/rest states suggests that they do not directly echo enhanced hippocampal sequence activity during sleep, questioning their role in consolidation. During outward runs, sequences seem to reflect the current behavioral state of the animal in the task and in space, corroborating that mPFC encodes task phase not only on the single cell level but also on the population level [Muysers et al., 2025]. These outward sequences could, thus, simply arise from trials with high firing rates. More interestingly though, is our finding on replay of GSs and TSs in the reward area. In contrast to the hippocampus where sequences of place cell activity are expressed in reversed order at the end of the track but in forward order at the beginning of the track [Diba & Buzsaki, 2007; Foster & Wilson, 2006] we found exclusively forward replay (**Fig. 5H**). Together with our finding of increased burst activity during error trials, this suggests that sub-second mPFC sequences may be associated with the evaluation of behavior. In addition to GSs and TSs, we found that most of the recurring sequences are not related to behavior (not SI), even during outward runs. Ongoing mPFC activity hence may reflect latent variables beyond experimental control.

The distinctiveness of hippocampal and PFC sequences has been emphasized previously [Kaefer et al., 2020; Nardin et al. 2023]. It is broadly assumed that mPFC sequences positively correlate with behavioral performance [Kaefer et al., 2020; Tang et al. 2021] – consistent with our observed establishment of stable task sequences after learning – and that they are associated with trajectory reactivations [Tang et al. 2021; Nardin et al., 2023], which is consistent with our finding of enhanced trajectory indices in the reward area (**Fig. 5H**). However, the suggestion that mPFC sequences may also support planning [Tang et al., 2021], could not be confirmed by our work as sequences in the odor sampling phase were absent (**Fig. 5H**). The difficulty of interpreting planning-associated activity may also partly result from the many activity bursts that are not task-related (**Fig. 5C**) and the relatively smaller amount of GCs and TCs being active in the odor sampling phase during population bursts (**Fig. 5D**). Moreover, planning related sequences may be masked by activity related to active sensing. Additionally, the 1-p calcium signal may simply not be able to detect planning-associated activity in case it only relies on weak activity modulations.

In cognitive neurosciences, the prefrontal cortex is often assumed to support flexible behavior by schemas, i.e., prototypic behavioral patterns that can be applied under variable conditions that share the same important contextual aspect. PFC neural activity in rodents has also been linked to schemas when task structure remained stable across spatial locations [El-Gaby et al., 2024]. The use of schema-like activity during learning of a novel task, however, is only little investigated. A small sample of cells has previously been identified to maintain task structure during rule switching [Reinert et al., 2021]. While the limited subset of GCs we found to translate their place fields into a generalized task field over learning could be interpreted as a schema of the spatial environment, task-related activity mostly developed independently from previous spatial representations. We thus hypothesize that in order to transfer a larger degree of schema-related representations to a new task, higher behavioral similarity between the tasks is required.

Methods

Experimental data was mostly published previously in Muysers et al. (2024) and Muysers et al (2025). Data derived from periods of sleep and habituation to the arena are previously unpublished. Detailed description of experimental methods can be found in the original publication. The next paragraph only provides a brief summary.

Experimental Methods and Data Preprocessing

Data was derived from 4 Thy1-GcaMP6f mice (Jackson Labs #025393; 2 female & 2 male). Animals were 11 weeks (75-78 days) old upon lens implantation and 16 weeks upon the beginning of the handling. The analyzed animals represented the subset of the previously published animals (Muysers et al., 2024, 2025) for which sufficiently many cells could be tracked throughout the experiment (including sleep and arena) to perform longitudinal sequence clustering.

Animals were first habituated to the experimenter and the experimental room for at least three days. The animals were then exposed to the behavioral arena (M-shaped maze) for three days (habituation to arena). The two different odors (vanilla; coconut) and the reward were introduced subsequently and the animals were exposed for a week to ‚forced‘ trials: The odor was presented at the center arm of the arena and only the correct door to the respective side-arm was opened, a reward was consequently given at the end. Thereafter, the “free choice” trials started, where the odor was presented at the center and both side-arms were accessible for free choice. Animals were trained for two months. In this time, all four mice learned the task (≥70% correct across three consecutive days).

Behavioral epochs (sampling, center, side, reward, all for left and right respectively) of the task were extracted, based on xy-thresholds of arena locations obtained from camera tracking.

Integrated GRIN lenses of ø 1mm (Inscopix #1050-004637) were used to gain optical access to the prefrontal cortex via a ø ~1mm craniotomy above the mPFC (centering on 1.7 mm anterior and 0.6 mm lateral of bregma).

Neuronal activity was recorded when the animals were first exposed to the behavioral arena and throughout all training sessions. For analysis we focused on the recordings of three days freely moving in the arena, four days of learning (~50% correct trials) and four days after reaching the criterion (“learned”). Additionally, animals were recorded during sleep in their homecage before and after learning of the task.

The open source Python toolbox Caiman was used to identify neurons in the recorded 1-photon videos. Extracted components were manually checked (*https://github.com/chenhungling/CaimanGUI*) before further analysis.

Calcium traces, as extracted with Caiman, were first corrected for slow drifts using a running percentile filter (10th percentile, 30s window). The mean (baseline) and standard deviation (σ) were then calculated iteratively. In each iteration, signal above 3σ was excluded and baseline and σ was calculated again until the relative change in σ was smaller than 0.1%. The traces were baseline-subtracted and normalized by the respective σ.

Significant transients were detected by keeping signal >3σ and a minimum duration of 0.2 s. The rest of the signal was set to 0. CellReg was used to track cells across sessions (Sheintuch et al., 2017).

Reactivation analysis

We followed Peyrache et al., (2010) and computed reactivation strength R_l of a population pattern b_l evoked by z-scored activity population vectors z (during sampling or reward) according to $R_l = (b_l^T z)^2$, with b_l denoting the l-th significant eigenvector of the n x n covariance matrix of the z-scored activity vectors during the outward (task) phase (with n denoting the number of simultaneously recorded neurons). Time binning was extended to 250ms and only those session were used in which the total duration of outward phase for each trial type (left/right) was exceeding 12.5s. Significance of eigenvectors was assessed via the Marchenko-Pastur bound $(1+(n/T)^{1/2})^2$, with T denoting the number time samples. Total reactivation R was then calculated as the weighted sum of all time averaged R_l , weighted with the corresponding eigenvalue of pattern b_l.

Burst detection

The population activity for each recording session was calculated by summing the neuronal activity across neurons at each time point. Subsequently, for each animal, the population activity from all sessions was concatenated to estimate the overall activity.

Burst events were identified when population activity exceeded a threshold of 0.5 standard deviations (σ). Neural activity from active cells within a 500 ms window, centered on each burst, was extracted and cells were sorted based on the center of mass time point of the calcium trace. For each population burst, the sorted indices of these cells were compiled into individual sequence vectors. Only sequences comprising at least five cells were included in subsequent analyses. Using the center of mass inherently normalizes for heterogeneous firing rates among neurons, a factor that can lead to spurious sequence order in electrophysiological recording, because high firing neurons would always fire earlier.

Clustering sequences

To quantify the similarity between sequence pairs, Spearman's rank correlation coefficients were calculated based on the ordinal ranks of neuron activations in each pair of bursts. Only the l neurons active in both sequences of a pair were taken into account. By construction Spearman's correlations are symmetric. In order to check for significance of sequence similarity, we followed the approach detailed in (Chenani et al., 2019): Correlations values were z-scored according to the distribution of correlation coefficients derived from random sequences of same length l. Only pairs with significant z score were considered further. The resulting binary matrix (1: significant, 0: non-significant) was then used for agglomerative hierarchical clustering with Ward's linkage method, producing discrete sequence clusters.

Iterative merging of cluster templates

After initial clustering of sequences and construction of a template for each cluster by averaging its member bursts, we quantified cluster similarity using the procedure described above for individual sequences. At each iteration we identified the cluster pairs with significant similarity and merged their underlying members if the z-score exceeded a fixed threshold ($z > 2.5$). Following each merge, cluster templates and similarity matrix were recomputed, and the procedure was repeated until no pair satisfied the merging criterion. The workflow for cluster identification is summarized in **Figure 6**.

Occupancy

Occupancy was calculated by first generating a histogram of the animal's position. The counts in each bin of the histogram were multiplied by the time spent in the corresponding bin, converting position counts into occupancy values per bin. To smooth the occupancy data, a Gaussian filter with sigma = 5 cm was applied.

Rate map

The smoothed transient histogram was divided by the occupancy to obtain the firing rates for each position bin.

Spatial information

Spatial information was calculated using the rate maps of neurons to identify place cells:

$$I = \sum_x P(x) \cdot r(x) \cdot \log_2((r(x) + \varepsilon) / (r_0 + \varepsilon))$$

Where r(x) is the firing rate in a spatial bin x, P(x) is the probability of the animal being in bin x, $r_0$is the overall mean firing rate, and ε is a small constant (1e-10) to prevent division by zero.

To identify place cells with significant spatial information, we performed circular shuffling of transient times within each trial period and calculated the spatial information for the shuffled rate maps. This process was repeated 500 times, generating 500 shuffled datasets. The spatial information of the original data was then compared to the shuffled distributions. Neurons with spatial information significantly greater than the shuffled data were classified as spatially informative (SI) cells.

Bayesian decoder

A Bayesian decoder was employed to calculate the posterior probabilities of all position bins. This method assumes that transient counts follow a Poisson distribution and combines transient counts and rates to compute the likelihood of the animal being at each position bin, given by:

$$\log L(y) = \sum_{i=1}^{N} [\, k_i \log(r_i(y) \cdot \Delta t) - r_i(y) \cdot \Delta t - \log(k_i!) \,]$$

Where L(y) is the likelihood of the animal being at position y, N is the total number of neurons, k_i is the count of transients of neuron i, $r_i(y)$ is the mean transient rate of neuron i at position y, and Δt is the decoding time bin.

The posterior probability P(y) is obtained by normalizing the likelihoods across all position bins:

$$P(y) = L(y) / \sum_{y'} L(y')$$

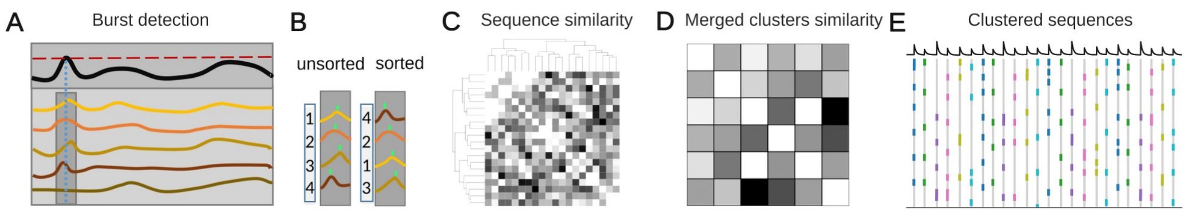


Figure 6.

Schematized workflow of sequence detection.

(A) For five example cells (calcium traces in colored lines) are summed to yield the population activity (black line). Whenever the population trace exceeds a threshold (red dashed line), a 500 ms second around the peak is considered a population burst (dark grey rectangle). (B) Single-cell activity within the 500 ms window is used to calculate center of mass for every cell. For every burst cells are reordered according to their center of mass. The resulting ordered indices are referred to as "sequence". (C) A sequence similarity matrix is computed from the z-scored Spearman correlation between two sequences according to (Chenani et al., 2019); The binarized matrix (1: significant z value, 0: non significant z-value) is subject to hierarchical clustering. (D) Similarity z-scores are again computed for the all pairs of cluster templates (mean sequences in a cluster) and the clustering is iteratively repeated until merging criteria are no longer fulfilled. (E) Sorted sequences, with colors indicating cluster membership

The estimated position of the animal is determined as the position ŷ that maximizes the posterior probability $\hat{y} = \mathrm{argmax}_y\, P(y)$.

Position estimation

To estimate the animal's position, peaks of transient calcium traces were detected for each neuron. Peak detection was performed using the find_peaks function from Python's SciPy library, with the parameters set to height=0, width= 50ms, and distance = 100 ms. The height=0 parameter ensured that all peaks above zero were considered, width specified the minimum width of peaks to be detected, and distance set the minimum required distance between consecutive peaks to avoid detecting closely spaced events as separate peaks.

A Bayesian decoder was trained using the detected peaks that were not part of bursts or their neighboring points. The decoder was then tested on the time points where bursts occurred to estimate the animal's position.

Estimate position using burst data

To estimate the animal's position using burst signals of sequences, a Bayesian decoder was trained using rate maps from outward runs as the training data. Time points when sequences were detected were excluded from the training data to ensure that the decoder's training was not influenced by these events. The decoder was then tested exclusively on the 500 ms burst signals centered around the detected sequences. Event times of bursts during the sampling, outward, and reward periods were used as test data. The animal's position during these times was estimated by taking the maximum likelihood of the posterior probability distribution provided by the Bayesian decoder.

Subsampling

To account for the unequal number of trials between the learning and learned conditions, the number of sequences in the learned condition was equalized by subsampling. The learned data were subsampled to match the number of sequences in the learning condition. This process was repeated 50 times to ensure robustness. Clustering was then performed independently on each of the 50 subsampled datasets.

Kullback-Leibler (KL) Divergence

To quantify the differences in cluster distributions between the learning and learned conditions, the Kullback-Leibler (KL) divergence was applied. The KL divergence is defined as $DKL(P \,||\, Q) = \Sigma_i\, P(i) \log(P(i) / Q(i))$, where P(i) and Q(i) represent the probabilities of sequence cluster i during the learning and learned conditions, respectively.

The KL divergence was calculated for each of the 50 subsampled datasets, resulting in a distribution of KL values derived from the original data. To assess significance, a null distribution was generated by randomly assigning sequences to the learning and learned conditions and computing the KL divergence for 50 such shuffled datasets. The KL divergence from the original data was considered significant if it exceeded the 95th percentile of the null distribution.

Assigning sleep and arena sequences to clusters

Assignment of sleep, arena, and task sequences to clusters was performed by calculating the rank order correlation between each sequence and all cluster templates. Each sequence was then assigned to the cluster with the highest similarity score based on the computed correlations.

Cluster-to-Cell Distance

The distances between cells and clusters were calculated by measuring the differences between the peak positions of the cells' rate maps and the sequence cluster templates. Distances were histogrammed for two conditions: either cell and sequence cluster are of same type (task arm, task stem, place arm, place stem) or including all cells. The two resulting distributions of distances were compared using a two-sided Mann–Whitney U test.

Task Cells and Goal arm Cells

To classify SI cells as goal arm selective cells or generalized task phase selective cells based on their firing patterns, rate maps during both left and right runs were analyzed. Cells were first sorted according to their rate maps from left runs. The analysis was then repeated, sorting cells according to their rate maps from the right run. For both sorting conditions, correlation coefficients were computed between the sorted rate maps of left and right runs for each region. To assess significance, a null distribution was generated by computing correlations between the rate maps of one condition and cell-index shuffled rate maps of the other condition. Cells with correlation coefficients exceeding the 95th percentile of this null distribution were classified as significant TCs, indicating similar activity during left and right runs. Conversely, cells with lower correlations were classified as GCs.

The rate maps were divided into two regions: the stem (linearized position 0–0.5) and the arms (0.5–1). Hence, the analysis identified four cell types: GCs in the arms (GC-arm), GCs in the stem (GC-stem), task cells in the arms (TC-arm), and task cells in the stem (TC-stem).

Task Sequences and Goal arm Sequences

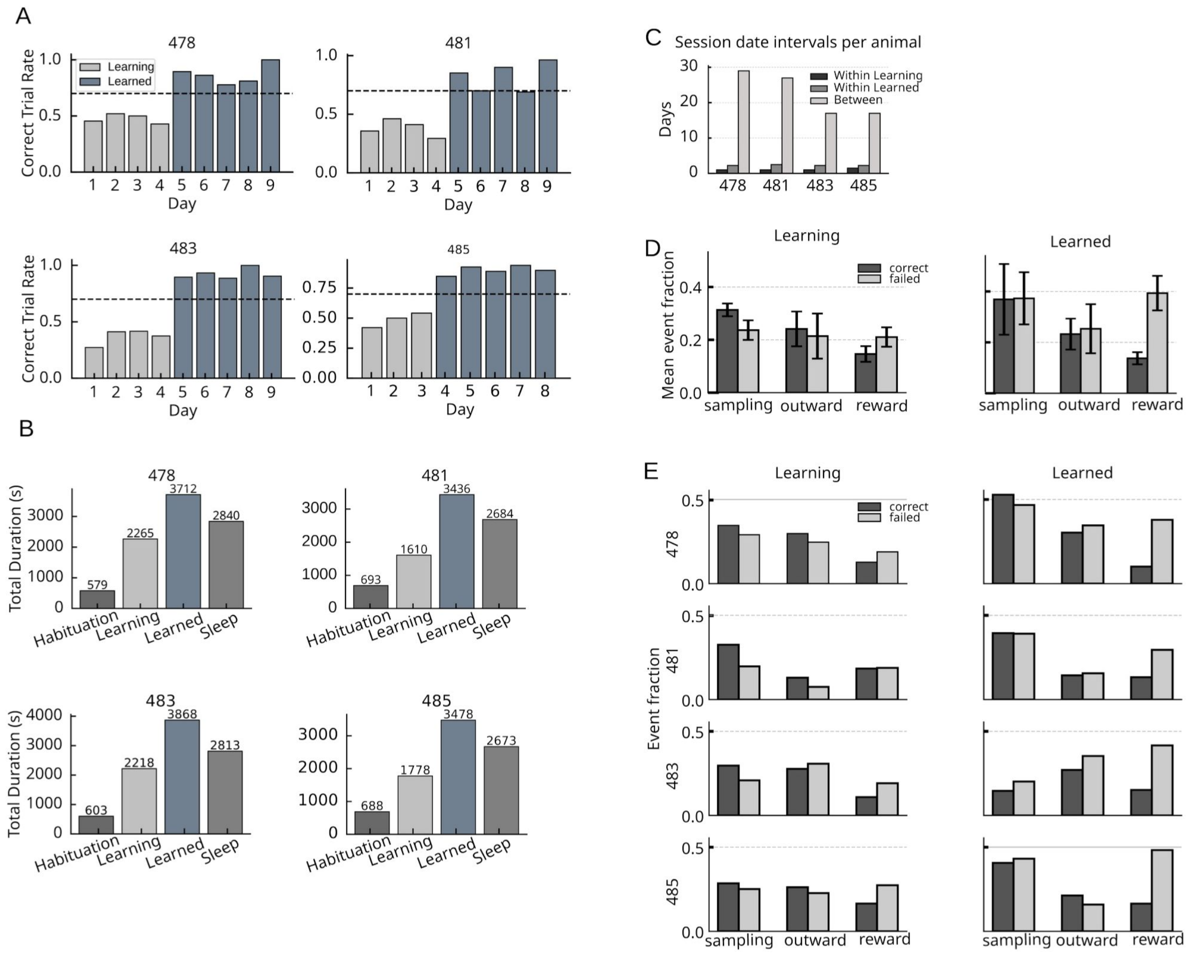
Criteria for defining GCs and TCs were also applied to categorize clusters based on their firing patterns and spatial activity into goal arm selective sequence clusters (GS) and generalized task phase sequence clusters (TS): Firing rate maps were computed for each cluster. Only clusters were further considered for which the at least one of the firing rate maps (Right/Left) had a significant SI. Rate maps were then correlated between left and right trials. If the correlation coefficient exceeded the 95 percentile of a cell index shuffle the clusters was considered a TS, otherwise a GS. The analysis was restricted to clusters containing at least five sequences.

*Stability of place field*s

To assess the stability of firing fields after learning, we compared the rate maps of SI cells between learning and learned condition. For each cell, the rate map of one condition was circularly shifted, and the correlation between the shifted and original maps was computed. If the maximum correlation occurred within a shift range of ±0.1, the cell was considered stable.

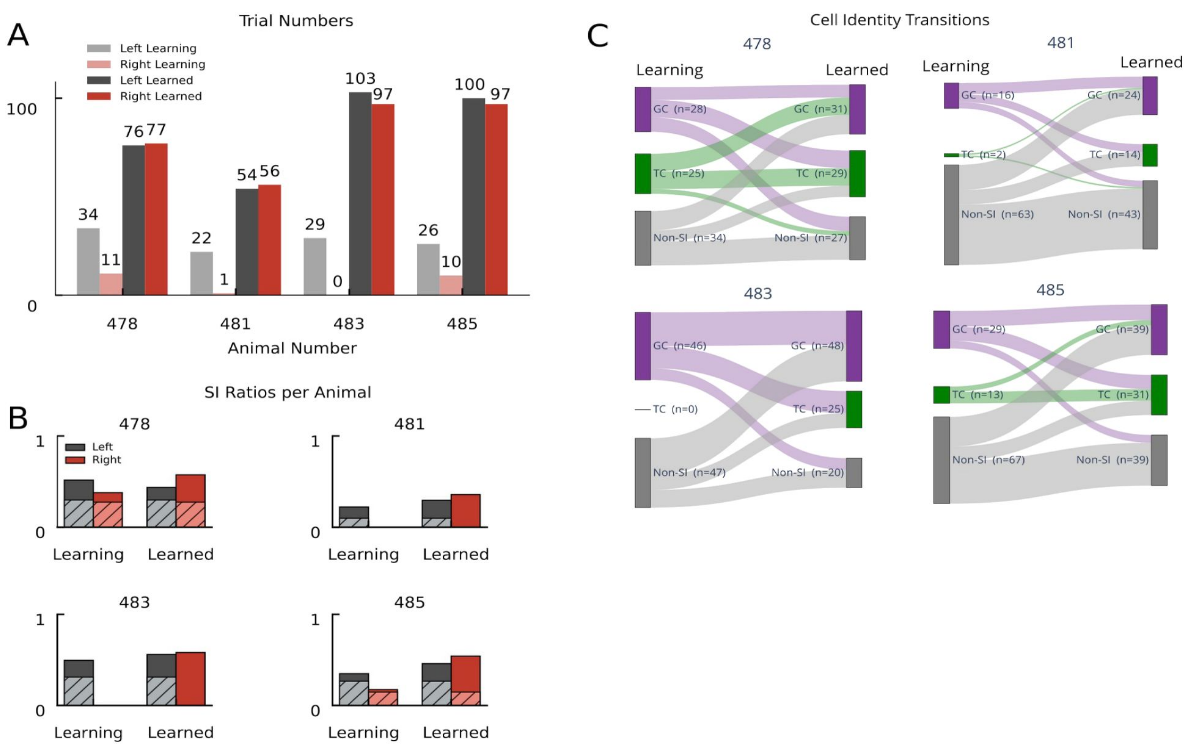
Data and Code availability

Analysis code is available from the GitHub-Repository *https://github.com/HamedShabani/PFC_sequence_analysis*. The datasets that went into the analyses are available under *https://doi.org/10.12751/g-node.b8k1yn*. Part of the data is overlapping with the repository of a previous publication (*https://doi.org/10.5281/zenodo.10528243*).



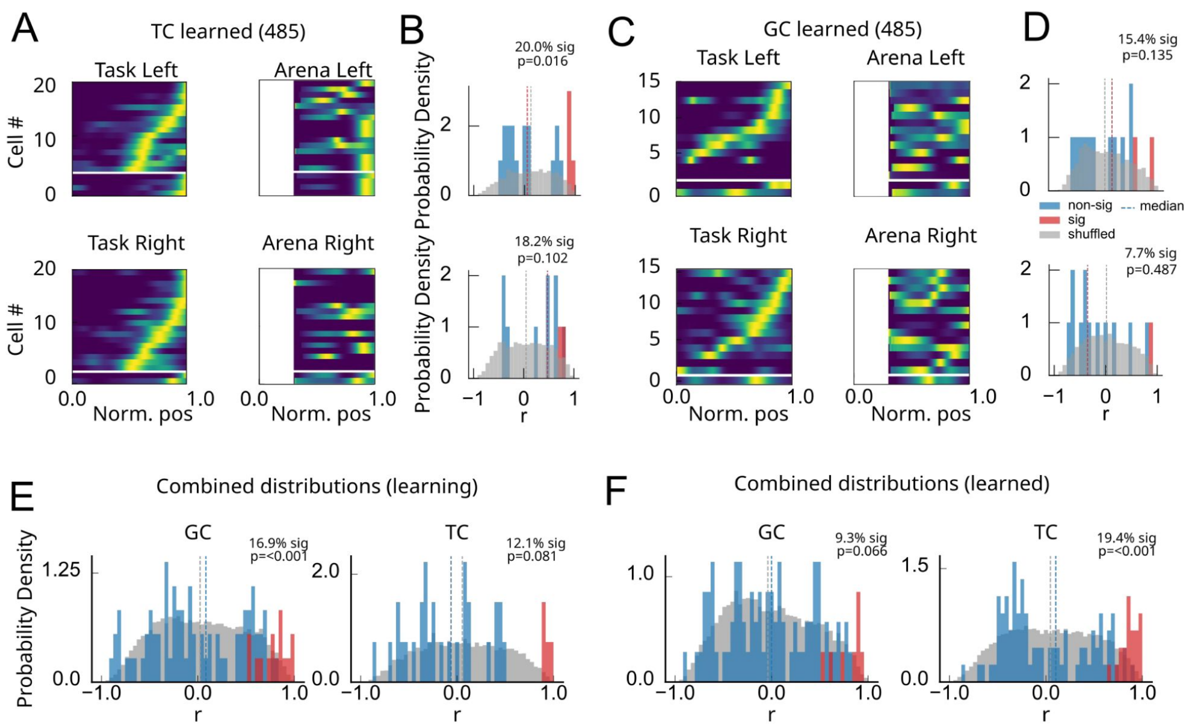
Supplementary Figure 1.

(A) Ratios of correct trials the animals included in the study (487, 418, 483, 485). The dashed horizontal line at 0.70 indicates the criterion used to distinguish between learning and learned sessions. (B) Total duration of recordings spent in behavioral epochs for individual animals during habituation (arena), task learning, task learned, and sleep. (C) Intersession intervals for each animal. (D) Fractions of time covered by population bursts with sequences from identified clusters. (E) Same as D separated for each animal.



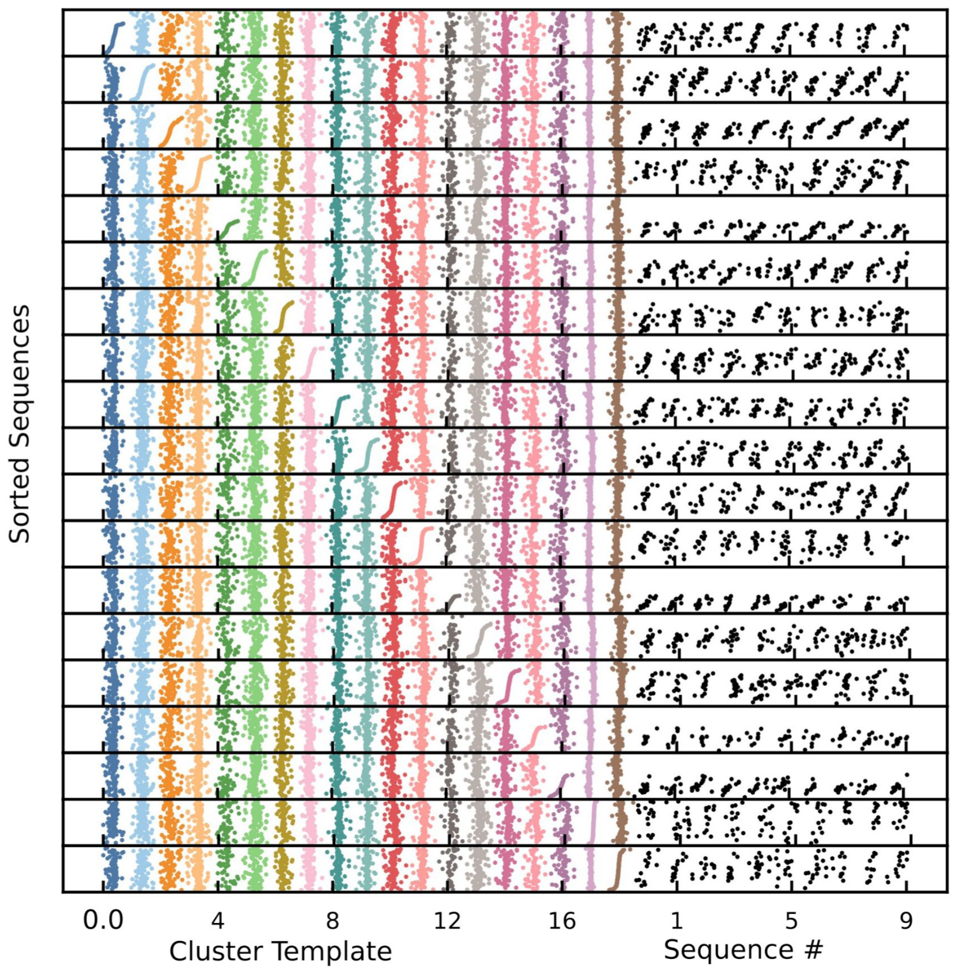
Supplement 1 to Figure 2.

(A) Cumulative correct trial numbers of individual animals (478, 481, 483, 485) for left and right runs during learning and after learning (learned). (B) Fraction of SI cells in the learning and learned condition. Same as **figure 2B**, but shown separately for individual animals. Only correct trials are shown and used in the analysis. (C) Transitions of cell identity between learning and learned condition, presented for individual animals.



Supplement 2 to Figure 2.

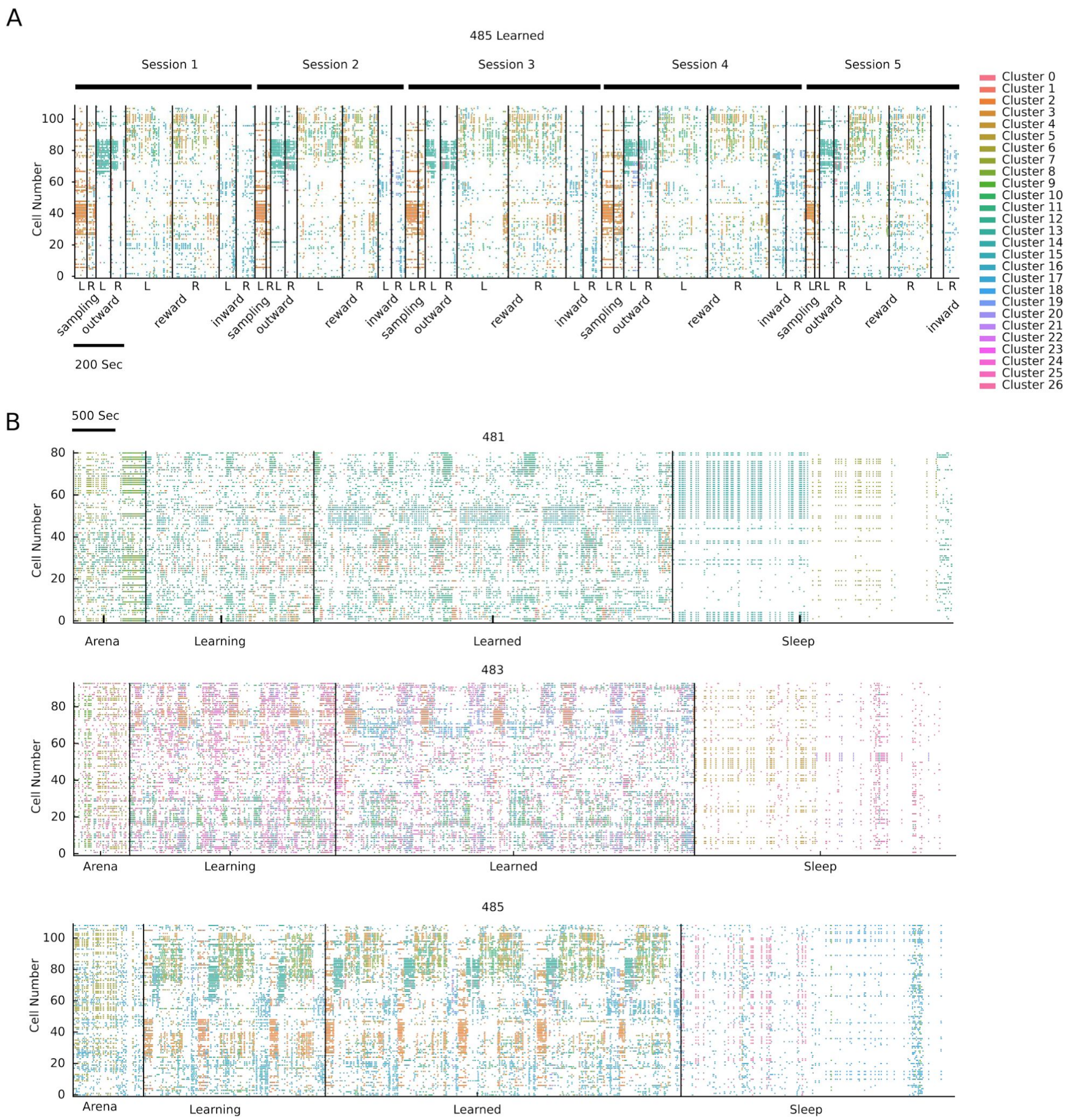
(A) Rate maps of TCs from an example animal (485) during proficient task performance (left) and arena habituation (without task; right). Because of the exploration pattern of the animals during arena habituation we could only compare the data mostly from the side arms. Stem areas with too little occupancy were excluded. Lower rows (separated by white stripe) depict cells with rate maps significantly similar between leaned condition an habituation. (B) Distribution of Pearson's correlation between place maps (blue/red). Red bars indicate significance with respect to circular shuffling of rate maps. Grey depicts shuffle distribution. P-values report significance of fractions of cells with significantly correlated rate maps (binomial test, 0.05). (C,D) Same as A,B for GCs. (E,F) Results cumulated over animals for the learning (E) and the learned (F) condition.



Supplement 1 to Figure 4.

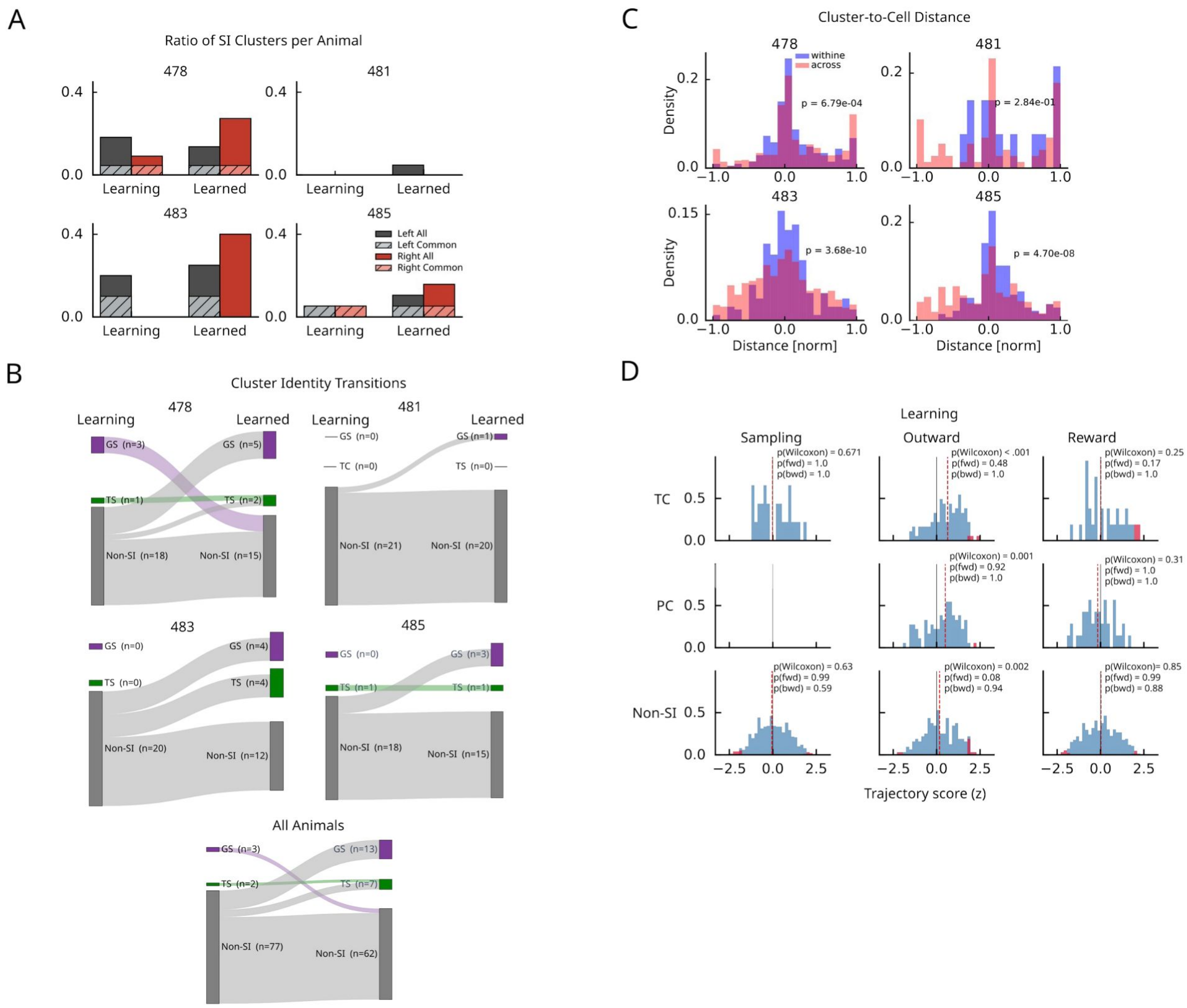
Complete set of sequence cluster templates (colored) obtained via hierarchical clustering sorted according to cluster (see Fig. 4C).

Black sequences are representative cluster members of the cluster for which the cell index sorting was made in the panel.



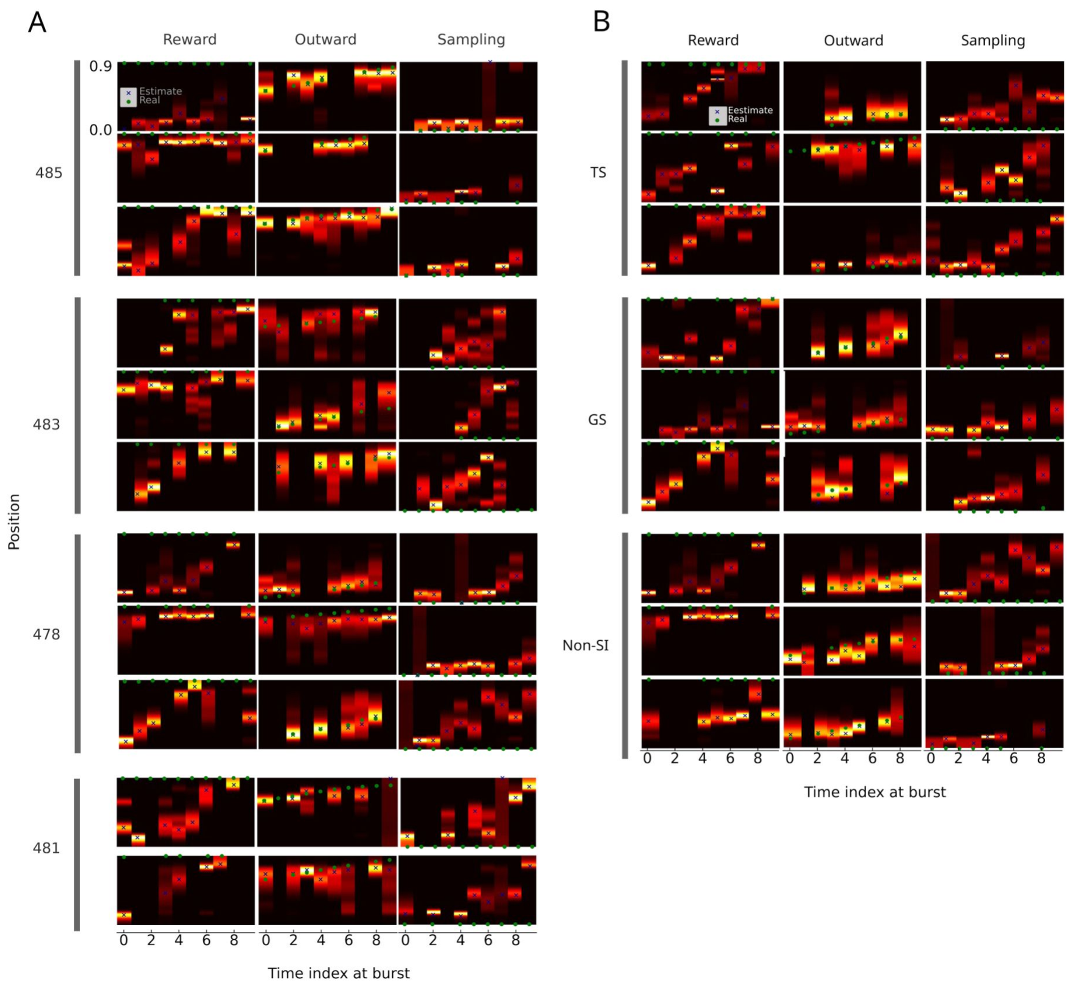
Supplement 2 to Figure 4.

(A) Rastermap visualization [Stringer et al., 2019] (embedding dimension = 1, n_X=60) of clustered sequences for an example dataset from a single animal for all sessions in the learned condition concatenated. Each point represents the time a cell had a transient within a sequence, with colors indicating the sequence cluster. (B) Rastermap visualization of data from three more individual animals (482, 483, 485), including habituation and sleep sessions as in **Fig. 4** D. Sleep sessions were dispersed throughout but shown here as a concatenated block for visualization purposes.



Supplement 1 to Figure 5.

(A) Fraction of SI sequence clusters (relative to all clusters) for individual animals (478, 482, 483, 485) in the learning and learned condition. Hatched areas indicate proportions significant in both conditions. (B) Transitions of cluster identity between learning and learned condition, presented for individual animals and for the merged data of all animals. (C) Histogram of distances between field peaks of the cluster and the field peaks of the contributing cells. Same as **Figure 4E**, but shown separately for individual animals. (D) Distribution of behavioral correlations during learning. Same as **Fig. 4H** for the sessions in the learning condition. Only outward phase shows median significantly above zero.



Supplement 2 to Figure 5.

(A,B) Additional examples of decoding during bursts from different behavioral phases and cluster types (Same as **Fig. 5G**), ordered according to animal (A) or sequence type (B).

Acknowledgements

This work was funded by the German Research Association (DFG) under grant numbers LE2250/20-1 (C.L., FOR5159), BA 1582/16-1 (M.B. FOR5159), BA 1582/24-1 (M.B.), and SA 3609/2-1 (J-F.S.FOR 5159).

Additional information

Funding

Deutsche Forschungsgemeinschaft (LE2250/20-1)

Deutsche Forschungsgemeinschaft (BA1582/16-1)

Deutsche Forschungsgemeinschaft (BA1582/24-1)

Deutsche Forschungsgemeinschaft (SA3609/2-1)

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Reviewer #1 (Public review):

Summary:

The authors use longitudinal in vivo 1-photon calcium recordings in mouse prefrontal cortex throughout the learning of an odor-guided spatial memory task, with the goal of examining the development of task-related prefrontal representations over the course of learning in different task stages and during sleep sessions. They report replication of their previous results, Muysers et al. 2025, that task and representations in prefrontal cortex arise de novo after learning, comprising of goal selective cells that fire selectively for left or right goals during the spatial working memory component of the task, and generalized task phase selective cells that fire equivalently in the same place irrespective of goal, together comprising task-informative cells. The number of task-informative cells increases over learning, and covariance structure changes resulting in increased sequential activation in the learned condition, but with limited functional relevance to task representation. Finally, the authors report that similar to hippocampal trajectory replay, prefrontal sequences are replayed at reward locations.

Strengths:

The major strength of the study is the use of longitudinal recordings, allowing identification of task-related activity in the prefrontal cortex that emerges de novo after learning, and identification of sub-second sequences at reward wells.

Comments on revisions:

The authors have added additional analyses and clarifications that increase the strength of evidence, especially quantification of functional classes of cells using longitudinal calcium recordings in prefrontal cortex during learning of an odor cue guided task, and quantification of prefrontal sequences.

There are a few remaining issues:

(1) The manuscript quantifies changes over learning in prefrontal goal-selective cells (equated to "splitter" place cells in hippocampus) and task-phase selective cells (similar to non-splitter place cells that are not goal modulated). A subset of these task cells remain stable throughout learning, and are equated to schema representations in the study. In the memory literature, schemas are generally described as relational networks of abstract and generalized information, that enable adapting to novel context and inference by enabling retrieval of related information from previous contexts. The task-phase selective cells that stay stable throughout learning clearly will have a role in organizing task representations, but to this reviewer, denoting them as forming a schema is an unwarranted interpretation. By this definition, hippocampal non-splitter place cells that emerge early in learning and are stable over days would also form a schema. Therefore, schema notation cannot just be based on stability, it requires further evidence of abstraction such as cross-condition generalization.

(2) The quantification of prefrontal replay sequences during reward is useful, but it is still unconvincing that the distinction between existence of sequences in the odor sampling phase and reward phase is not trivially expected based on prior literature. This is odor guided task, not a spatial exploration task with no cues, and it is very well-established (as noted in citations in the previous review) that during odor sampling, animals' will sniff in an exploratory stage, resulting in strong beta and respiratory rhythms in prefrontal cortex. Not having LFP recordings in this task does not preclude considering prior literature that clearly shows that odor sampling results in a unique internal state network state, when animals are retrieving the odor-associated goal, vastly different from a reward sampling phase. The authors argue that this is not trivial since they see some sequences during sampling, although they also argue the opposite in response to a question from Reviewer 2 about shuffling controls for sequences, that 'not' seeing these sequences in the sampling phase is an internal control. The bigger issue here is equating these sequences during sampling to replay/ preplay or reactivation sequences similar to the reward phase, since the prefrontal network dynamics are engaged in odor-driven retrieval of associated goals during sampling, as has been shown in previous studies.

https://doi.org/10.7554/eLife.106981.2.sa3

Reviewer #2 (Public review):

Summary:

The first part of the manuscript quantifies the proportion of goal-arm specific and task-phase specific cells during the learning and learned conditions and similar to their previously published Muysers et al., 2025 paper find that the task-phase coding cells (Muysers et al. call them path equivalent cells) increase in the learned condition. However, compared to the Muysers et al. 2025 paper, this work quantifies the proportion of cells that change coding type across learning and learned conditions. The second part of the paper reports firing sequences using a sequence similarity clustering-based method that the group developed previously and applied to hippocampal data in the past.

Strengths:

Identifying sequences by a clustering method in which sequence patterns of individual events are compared is an interesting idea.

Weaknesses:

Further controls are needed to validate the results.

Comments on revisions:

Further changes are needed to improve the description of the methods and the discussion needs to be extended to contrast the results with previously published results of the group. Some control figures would also be needed to quantitatively demonstrate, across the entire dataset, that sequence detection did not identify random events as sequences, even if the detection method was designed to exclude such sequences. For example, showing that sequences are not detected in randomised data with the current method would better convince readers of the method's validity.

Although differences in the classification scheme relative to the Muysers et al. (2025) paper have been explained, the similarity (perhaps equivalence of results) is not sufficiently acknowledged - e.g., at the beginning of the discussion.

Although the control of spurious sequences may have been built into the method, this is not sufficiently explained in the method. It is also not clear what kind of randomization was performed. Importantly, I do not see a quantification that shows that the detected sequences are significantly better than the sequence quality measure on randomized events. Or that randomized data do not lead to sequence clusters. Also, it is still not clear how the number of clusters was established. I understand that the previously published paper may have covered these questions; these should be explained here as well. Also, the sequence similarity description is still confusing in the method; please correct this sentence "Only the l neurons active in both sequences of a pair were taken into account. "

https://doi.org/10.7554/eLife.106981.2.sa2

Reviewer #3 (Public review):

In the study the authors performed longitudinal 1P calcium imaging of mouse mPFC across 8 weeks during learning of an olfactory-guided task, including habituation, training, and sleep periods. The authors' goal was to determine how the mPFC representation of the task changed and what aspects of activity emerged between the learning and the learned conditions of the task. The task had 3 arms. Odor was sampled at the end of the middle arm (named the "Sample" period). The animal then needed to run to one of the two other arms (R or L) based on the odor. The whole period until they reached the end of one of the choice arms was the "Outward" period. The time at the reward end was the "Reward" period. They noted several changes from the learning condition to the learned condition:

(1) They classified cells in a few ways. First each cell was classified as SI (spatially informative) if it had significantly more spatial information than shuffled activity, and ~50% of cells ended up being SI cells. Then among the SI cells they classified a cell as a TC (task cell) if it had statistically similar activity maps for R versus L arms, and a GC (goal arm cell) otherwise. Note that there are 4 kinds of these cells: outer arm TCs and GCs and middle arm TCs and GCs (with middle arm GCs essentially being like "splitter cells" since they are not similarly active in the middle arm for R versus L trials). There was an increase in TCs from the learning to the learned condition sessions. They also note the sources of these TCs (some came from GCs, others from non-SI cells).

(2) They analyze activity sequences across cells. They extracted 500 ms duration bursts (defined as periods of activity > 0.5 standard deviations over what I assume is the mean, which is a permissive threshold encompassing a significant fraction of the activity in non-sleep, non-habituation periods). They first noted that the resulting "Burst rates were significantly larger during behavioral epochs than during sleep and during periods of habituation to the arena" and "Moreover, burst rates during correct trials were significantly lower than during error trials". For the sequence analysis they only considered bursts consisting of at least 5 active cells. A cell's activity within the burst was set to the center of mass calcium activity. Then they took all the sequences from all learned and learning sessions together and hierarchically clustered them based on the Spearman's rank correlation between the order of activity in each pair of sequences (among the cells active in both). The iterative hierarchical clustering process produces groups (clusters) of sequences such that there are multiple repeats of sequences within a cluster. Different sequences are expressed across all the longitudinally recorded sessions. They noted "large differences of sequence activation between learning and learned condition, both in the spatial patterns (example animal in Fig. 4D) and the distribution of the sequences (Fig. 4D,E). Rastermap plots (Fig. 4D) also reveal little similarity of sequence expression between task and habituation or sleep condition." They also note the difference in the sequences between learning and learned condition was larger than the different between correct and error trials within each condition. They conclude that during task learning new representations are established, as

measured by the burst sequence content. They do additional analyses of the sequence clusters by assessing the spatial informativeness (SI) of each sequence cluster. Over learning they find an increase in clusters that are spatially informative (clusters that tend to occur in specific locations). Finally, they analyzed the SI clusters in a similar manner as SI cells and classified them as task phase selective sequences (TSs) and goal arm selective sequences (GSs) and did some further analysis. However, they themselves conclude that the frequency of TSs and GSs is limited because most sequence clusters were non-SI. In the discussion they say "In addition to GSs and TSs, we found that most of the recurring sequences are not related to behavior (not SI)".

(3) As an alternative to analyzing individual cells and sequences of individual cells, they then look for trajectory replay using Bayesian population decoding of location during bursts. They analyze TS bursts, GS bursts, and non-SI bursts. They say "we found correlations of decoded position with time bin (within a 500 ms burst) strongly exceeding chance level only during outward and reward phase, for both GSs and TSs (Fig 5H)." Fig5H shows distributions indicating statistically significant bias in the forward direction (using correlations of decoded location versus time bin across 10 bins of 50 ms each within each 500ms burst). They find that the Outward trajectories appear to reflect the actual trajectory during running itself, so are likely not replay. But the sequences at the Reward are replay as they do not reflect the current location. Furthermore, replay at the Reward is in the forward direction (unlike the reverse replay at Reward seen in the hippocampus) and this replay is only seen in the learned and not the learning condition. At the same time, they find that replay is not seen during odor Sampling, from which they conclude there is no evidence of replay used for planning. Instead they say the replay at the Reward could possibly be for evaluation during the Reward phase, though this would only be for the learned condition. They conclude "Together with our finding of strong changes in sequence expression after learning (Fig 4E) these findings suggest that a representation of task develops during learning".

This study provides valuable new information about the evolution of mPFC activity during the learning of a odor-based 2AFC T-maze-like task. They show convincing evidence of changes in single cell tuning, population sequences, and replay events. They also find novel forward replay at the Reward, and find that this is present only after the animal learned the task. In the discussion the authors note "the present study, to our knowledge, identified for the first time fast recurring neural sequence activity from 1-p calcium data, based on correlation analysis". Overall, they find a substantial amount of change among the features they analyzed and according to their methods, though they note a small amount of activity was preserved through learning.

One comment is that the threshold for extracting burst events (0.5 standard deviations, presumably above the mean) seems lower than what one usually sees as a threshold for population burst detection, and the authors show (in Supplementary Fig 1) that this means bursts cover ~20-40% of the data. However, it is potentially a strength of this work that their results are found by using this more permissive threshold.

https://doi.org/10.7554/eLife.106981.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *The study mainly replicates the authors' previously reported results about generalized and trajectory-specific coding of task structure by prefrontal neurons, and stable and changing representations over learning (Muysers et al., 2024, PMID: 38459033; Muysers*

> *et al., 2025, PMID: 40057953), although there are useful results about changes in goal-selective and taskphase selective cells over learning. There are basic shortcomings in the scientific premise of two new points in this manuscript, that of the contribution of pre-existing spatial representations, and the role of replay sequences in the prefrontal cortex, both of which cannot be adequately tested in this experimental design.*

We agree with the reviewer that we have not made sufficiently clear which aspects of our paper add to previous publications. We have now better explained methodological differences.

Also, we agree that our very general statements on pre-existing spatial representations in the introduction and abstract in the previous manuscript were not properly followed up in the Results section. In the revision, the respective statements are clarified, and we also added analysis of a further control condition (see response to A), which shows that particularly a subset of task cells maintains there firing fields from an early habituation period, arguing that, while the population representation of the task largely develops during learning, there exists a scaffold of small but significant amount of cells that could be interpreted as a schema.

We also further clarified our view on replay sequences in the prefrontal cortex (see response to B). Particularly, we are grateful to the reviewer for the suggestion to also include other reactivation analysis which led to new results presented in new Figure 3.

> *[A] The study denotes neurons that show precise spatial firing equivalently irrespective of goal, as generalized task representations, and uses this as a means to testing whether pre-existing spatial representations can contribute to task coding and learning. .... [I]n order to establish generalization for abstract task rules or cognitive flexibility, as motivated in the manuscript, there is a need to show that these neurons "generalize" not just to firing in the same position during learning of a given task... For an adequate test of pre-existing spatial structure, either a comparison task, as in the examples above, is needed, or at least a control task in which animals can run similar trajectories without the task contingencies. An unambiguous conclusion about pre-existing spatial structure is not possible without these controls.*

We thank the reviewer for this suggestion. We may, however, note that the previous manuscript did not make strong claims about pre-existing structures in the Results or Discussion. Also Schemas were only taken up as a discussion point. We nevertheless agree with the reviewer that assessment of the spatial prestructure requests further analysis. To address their point, we analyzed neuronal activity during the habituation phase before the start of task training, when the animals freely explored the same maze without any task contingency (animals explored mostly in the arms of the maze). We compared the place fields of neurons during this habituation period with their task-related activity. Consistent with the small overlap of firing rate maps between learning and learned phase, also this analysis revealed a small number of cells with significant correlations (up to 20% for task cells; a significant fraction according to a binomial test). The results are shown as a new Figure supplement to Figure 2.

> *[B] The scientific premise for the test of replay sequences is motivated using hippocampal activity in internally guided spatial working memory rule tasks [...] and applied here to prefrontal activity in a sensory-cue guided spatial memory task [...]. There are several issues with the conclusion in the manuscript that prefrontal replay sequences are involved in evaluating behavioral outcomes rather than planning future outcomes.*

We agree with the reviewer that preplay in Hippocampus and mPFC are distinct. We further emphasized this distinctiveness in the respective paragraph in the discussion (see response to B1).

> *[B. 1] First, odor sampling in odor-guided memory tasks is an active sensory processing state that leads to beta and other oscillations in olfactory regions, hippocampus, prefrontal cortex, and many other downstream networks [...]. This is an active sensory state, not conducive to internal replay sequences, unlike references used in this manuscript to motivate this analysis, which are hippocampal spatial memory studies with internally guided rather than sensory-cue guided decisions, where internal replay is seen during immobility at reward wells. These two states cannot be compared with the expectation of finding similar replay sequences, so it is trivially expected that internal replay sequences will not be seen during odor sampling.*

We agree with the reviewer that the sampling phase cannot be compared with the "preplay" state in the hippocampus. We have rewritten the manuscript in the results and discussion sections to clarify. We, however, disagree, that the absence of replay sequences in the mPFC 1P calcium data is trivial, since we actually do see many sequences during sampling (Fig 4E, Fig 4 suppl 2 A). These sequences are just not related to task activity and may e.g. reflect activity related to sensing, but do not contain information about goal arm.

> *[B. 2] Second, sequence replay is not the only signature of reactivation. Many studies have quantified prefrontal replay using template matching and reactivation strength metrics that do not involve sequences [...]. Third, previous studies have explicitly shown that prefrontal activity can be decoded during odor sampling to predict future spatial choices - this uses sensory-driven ensemble activity in prefrontal cortex and not replay, as odor sampling leads to sensory driven processing and recall rather than a reactivation state [..].*

We thank the reviewer for the suggestion to also perform reactivation analysis (Peyrache et al., 2009, 2010). The results are summarized in the new Figure 3. And show that indeed reactivation is stronger during the sampling phase and it is goal arm specific, arguing that sequence analysis extracts information (partly) complementary to rate covariance based analysis.

We hope to have convinced the reviewer that, together, the complementary results of reactivation an sequence analysis, as well as the ability to follow these measures over an extended period of time, gives unique insights far beyond the previous publications of these data sets. A consistent analysis of population representation, however, required some reanalyses of previous findings, since we only could focus on a limited number of animals and cells, for which tracking was possible over such a long period of time.

> ***Reviewer #2 (Public review):***
>
> *Further controls are needed to validate the results.*

We thank the reviewer for their generally supportive statements. The revised manuscript contains a number of controls in several new figure supplements.

> ***Reviewer #3 (Public review):***
>
> *[They] conclude that the frequency of TSs and GSs is limited (I believe because most sequence clusters were non-SI - the authors can verify this and write it in the text?). In the discussion, they say, "In addition to GSs and TSs, we found that most of the recurring sequences are not related to behavior".*

The reviewer is correct most clusters were not SI (Fig 5 A). We have added this information in the MS.

> *[...] They conclude "Together with our finding of strong changes in sequence expression after learning (Figure 3E) these findings suggest that a representation of task develops during learning, however, it does not reflect previous network structure." I am not sure what is meant here by the second part of this sentence (after "however ..."). Is it the idea that the replay represents network structure, and the lack of Reward replay in the learning condition means that the network structure must have been changed to get to the learned condition? Please clarify.*

The reviewer is correct in their assertion. We rewrote the sentence to clarify: "Together with our finding of strong changes in sequence expression after learning (now Fig 4E) these findings suggest that a representation of task develops during learning, however, it does not reflect sequence structure during learning and habituation".

> *(1) There are some statements that are not clear, such as at the end of the introduction, where the authors write, "Both findings suggest that the mPFC task code is locally established during learning." What is the reasoning behind the "locally established" statement? Couldn't the learning be happening in other areas and be inherited by the mPFC? Or are the authors assuming that newly appearing sequences within a 500-ms burst period must be due to local plasticity?*

We agree that the wording "local" can be misleading, we rephrased the corresponding sentences.

> *(2) The threshold for extracting burst events (0.5 standard deviations, presumably above the mean, but the authors should verify this) seems lower than what one usually sees as a threshold for population burst detection. What fraction of all data is covered by 500 ms periods around each such burst? However, it is potentially a strength of this work that their results are found by using this more permissive threshold.*

Since we work with a slow calcium signal, we cannot use as strict thresholds as usually employed using electrophysiology. In addition, our sequence detection approach adds a further level of strictness such that we only consider bursts with recurring sequence structure. In response to this reviewer's question, we have added quantification of the fraction of all data covered by 500 ms periods in Figure Supplement 1, panels D and E. Indeed we include a large fraction (20 to 40%) (except sleep and habituation), which is consistent with our interpretation that during the outward phase sequences mainly reflect task field firing.

> ***Reviewer #1(Recommendations for the Authors):***
>
> *It is possible that 1-photon recordings do not have the temporal resolution and information about oscillatory activity to enable these kinds of analyses. Therefore, an unambiguous conclusion about the existence and role of prefrontal reactivation is not possible in this experimental and analytical design.*

We indeed cannot extract information encoded in LFP oscillations from the calcium signal, we now mention the relation between LFP oscillations and olfaction-guided behaviors in the discussion (including the suggested references). However, our finding that sequence and covariance-based analysis yield partly complementary results argues that it indeed allows conclusion about the existence and role of prefrontal reactivation.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *The results of the Muysers et al. (2025) paper need to be discussed in detail and explain why the cell categorization is different, three groups of spatial cells vs two groups here. Also, explain in what aspect the major findings in this work go beyond what was shown in Figure 4 in that paper.*

The main goal of this paper was to explore sequence/replay like activity, which is not at all captured in the Muysers et al. 2025 paper. Because of this focus on sequences, we excluded the inward runs (from reward to sampling point) for better interpretability and thus ended up with only two types of cells. Muysers et al. included backward runs and could thereby also assess whether the place field remains in the outward and inward runs. We added this clarification in the Results section.

Regarding the reviewer's question regarding figure 4: Our task cells would largely overlap with the "path-equivalent cells" from Muysers et al. 2025 (albeit not taking into account inward runs). In this sense their finding that the share of path-equivalent cells increases with learning is consistent with our report of increasing fraction of task cells in Figure 2 C. Our Figure 2 adds that some task cells develop from previous goal cells with fields at the same location (generalizing). Moreover, we use spatial information as a criterion to identify TC and GCs, showing that a large fraction of cells actually is and remains spatially unselective. In Muysers et al. 2025 a statistical criterion was not applied on spatial selectivity but peak height, with fewer neurons failing this test. Moreover, we were analyzing only those cells trackable over the whole period. Despite all these methodological differences, the result of increasing the number of task/path-equivalent cells over learning was consistent. The main reason for recategorization of the cells in the present manuscript was to be able to meaningfully link them to sequence activity (Fig. 5E, F).

> *It is not clear from the description how the cell type transitions were quantified. Was the last learning day compared to the first learned day? Given that, particularly during learning, there are changes across days in the spatial representations according to Figure 2 of Muysers et al. (2025), this is the meaningful way to make the comparisons. Nevertheless, it is also not clear whether the daily variations within learning and learned conditions differ from the transition day, so without comparing these three conditions, it is hard to make a firm conclusion from examining only changes in the transition days.*

The analysis of cell type transitions was performed by pooling all learning sessions and comparing them with all learned sessions, without taking into account the chronological order of sessions within each category. This approach allowed us to identify broad changes associated with learning state. Figure supplement 1.C shows the session intervals per animal. We argue that the large interval between learning and learned session justifies this analysis approach.

> *Identifying sequences by a clustering method in which sequence patterns of individual events are compared is an interesting idea. Nevertheless, there is a danger, as with any clustering method, that data without clustering tendency could be artificially subdivided into clusters.*

In Figure 4.C, we show three example sequence cluster templates (colored) obtained via hierarchical clustering, along with representative member sequences (black) sorted by cluster membership. In response to this reviewer's comment, we now included a complete clustering result for one animal, including all sequence clusters and their member sequences. It is provided in Figure 4 supplement 1. This comprehensive visualization serves as an additional control, demonstrating that the clustering approach identifies consistent sequence patterns across the dataset.

> *Furthermore, it is possible that some cells at the edge of the cluster boundary may show a more similar sequence tendency to events detected at the overlapping border region of another cluster. Was this controlled for? It would be essential to show that events clustered together all show higher similarity to each other than to events in any other clusters.*

By default, the clusters are rejected if in the adjacency matrix of the graph constructed by significant motif similarity, the number of within cluster edges is smaller than the number of without cluster edges. In subsequent cluster merges the separation is increased since only those clusters are merged that show significant similarity. As a visual control, we monitor plots as shown in Figure 4 supplement 1. Sequence templates (color dot clouds) are supposed to show no serial correlation when ordered according to any one template other than its own. We have added more clarification to the Methods including a new Figure 6 illustrating the Method.

> *From the description, it was not clear how the sequence similarity was established between pairs of individual events. The only way I can see it is that the sequence (orders at which cells fire) is established with one event, and the rank order correlation is calculated with this order for the other event. However, in this case, distance A-B is not the same as distance B-A. Not sure how this is handled with the clustering procedure. Secondly, how the number of clusters is established in the hierarchical clustering procedure needs to be explained. Furthermore, from the method description, it is not clear how GS and TS sequences are identified. Can an event be classified as both a TS and GS event at the same time?*

The reviewer is correct in their assertion that we compute all pairwise rank order correlations (that are then subject to a statistical test detailed in the original method publication Chenani et al., 2019). By nature of the rankorder correlation the coefficients A-B and B-A are symmetric. This is now more carefully explained in the Methods.

> *Several control analyses are needed to show that the sequences detected reflect not random patterns but those that repeat at a higher than random chance. This requires, at the first step, to establish to what degree sequences are consistent within a cluster and to what degree individual events show a sequential firing tendency. And at the next stage, these need to be compared with randomised events in which spike timing of cells is jittered or spike identity is randomised, and show that these events result in poorer sequence tendency and less consistent clusters.*

The controls requested by the reviewer are already implemented in our Method (see original publication of the Method in Chenani et al., 2019). This is now made clearer in the Methods section.

> *Firing rate and place-related firing of cells alone could generate sequences even if cells otherwise fire independently from each other. In a similar manner, it was shown before that reactivation of waking cell assemblies could be seen in sleep, in which case firing rate differences across cells belonging to the same assembly could also generate sequential patterns without temporal coordination. Appropriate shuffling procedures need to be performed to exclude such scenarios.*

We are aware that the sequential firing in our data (particularly during the outward phase when the animal is performing the task), is most likely resulting from the correlations between rate maps and the animals trajectory. During the reward, this is less likely. An intrinsic control is that during sampling we do not see these sequences. Given the nature of the calcium signal, a direct connection to firing rate is not possible. However, we argue that using our center of mass-approach of the calcium trace effectively normalizes for firing rate effects. Shuffling dF/F amplitudes (as a proxy for firing rates) would thus have no effect on the center of mass sequences. We, however, consider this to be an important methodological difference between sequence analysis with spikes and Calcium signals and have added a related comment to the Methods part.

> *The past literature describing mPFC reactivation, replay, and sequences needs to be described, and findings of this work need to be appropriately acknowledged, and those findings compared with this work (starting with this work from 2007 PMID: 18006749). In the current reading, a novice reader of this field might conclude that this is the first work that identified relay and sequences in the mPFC.*

We would like to apologize that the manuscript evokes this impression. This was not our intention, in fact we have given strong emphasis on the Kaefer et al. paper in the Discussion. We have now added early references on PFC replay based on electro-physiological recordings in the Discussion section.

> *The analysis of Figure 4H is not sufficient to show that only forward sequences occur. If 50% are forward and 50% are reverse, the median is zero. Some of the presented histograms look like Gaussian distributions with SD=1, which would show that those events were not real sequences. It should be tested whether the distributions are significantly different from the expected Gaussian.*

We agree with the reviewer that we did not explicitly test for significance of individual replays, but only tested for the rightward shift of the median. We have now added these significance tests/p values in Figure 5) and indeed could show that none of the significant backward replays exceed the fraction expected by chance, whereas forward replay significantly exceeds chance levels only in the cases where the median had a significant right ward shift (except for non-SI clusters). We would like to thank the reviewer for this suggestion, which we think makes the analysis stronger.

> *Overall, the clarity of the text could be improved, and further examples of reactivated sequences should be shown, and the methods should be illustrated in the figures. At the current version, I fear that even readers in this field would give up on reading the current text given an insufficient level of clarity.*

We have included more examples of reactivated sequence (Suppl2 to Figure 5) and made extensive additions to the methods part. Particularly, we followed the reviewer's request for method illustration (new Figure 6).

> ***Reviewer #3 (Recommendations for the authors):***
>
> *My main comment here is for the authors to increase the clarity of the manuscript.[...] For instance, it was difficult to follow what was being done to determine TSs and GSs.*

We have made extensive additions to the Methods section including a new Figure 6 depicting the workflow of the sequence analysis in a schematic manner.

https://doi.org/10.7554/eLife.106981.2.sa0